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Forecasting the Success of Environmental and Sustainability Activities in International Development Using Language Models

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Abstract

Abstract in English

International aid and cooperation improve the lives of the poorest in developing countries and help safeguard the environment and promote sustainability. However, international aid activities sometimes fail to achieve their objectives. Few attempts have been made in the literature to create models that forecast the success of international aid activities, and none focus solely on environmental outcomes. This thesis produces a forecasting system for various metrics measuring the success of international aid activities at time of evaluation using data from the International Aid Transparency Initiative (IATI) database, combining classical statistical methods with modern language model techniques. Novel techniques are applied to improve forecasting skill, including using the reasoning and information-gathering abilities of large language models (LLMs) to improve forecasts, introducing LLM summaries of various dimensions of activity documents, and defining a novel narrative similarity grade benchmark. Forecasting success ratings on a scale from 1 to 6, narrative forecasts, cost-effectiveness forecasts, and finally true/false outcome tag forecasts are assessed. While no methods showed consistently high accuracy in forecasting overall success, statistical models could reliably rank above-chance which activities are more likely to succeed for activities with the same reporting organization and start year ("pairwise ranking"). The full forecasting system outperforms what could be extrapolated from the stated risks in activity documents alone for overall evaluations in tests against a held-out test set of 200 latest-starting evaluated activities in a dataset of 1181 environmental and sustainability improving activities restricted to 4 reporting organizations. Compared to the baseline of 51 % pairwise ranking for the risks extrapolation, the chosen statistical model with all input features reached 60 % [95% CI: 51 %, 66 %] on the test set. Across 14 binary outcome tags, statistical models achieve an average pairwise ranking of 60 % (range: 48 %–77 %) for the 45% of pairs with differing outcomes averaging two chosen methods on the test set. This work lays the foundation to improve decision making wherever there is a large collection of pre-intervention description documents and matched post-activity evaluation documents.

Abstract auf Deutsch

Internationale Hilfe und Zusammenarbeit verbessern das Leben der Ärmsten in Entwicklungsländern und tragen zum Schutz der Umwelt sowie zur Förderung der Nachhaltigkeit bei. Allerdings verfehlen internationale Hilfsmaßnahmen mitunter ihre Ziele. In der Fachliteratur finden sich bisher nur wenige Ansätze zur Entwicklung von Modellen zur Erfolgsprognose internationaler Hilfsmaßnahmen, und keiner davon konzentriert sich ausschließlich auf Umweltwirkungen. Diese Arbeit entwickelt auf Grundlage der Datenbank der International Aid Transparency Initiative (IATI) ein Prognosesystem für verschiedene Metriken zur Messung des Erfolgs internationaler Hilfsmaßnahmen zum Zeitpunkt der Evaluierung. Dabei werden klassische statistische Methoden mit modernen Techniken großer Sprachmodelle (Large Language Models, LLMs) kombiniert. Zur Verbesserung der Prognosegüte kommen neuartige Techniken zum Einsatz, darunter die Nutzung der Schlussfolgerungs- und Informationsbeschaffungsfähigkeiten großer Sprachmodelle, die Erstellung von LLM-Zusammenfassungen verschiedener Dimensionen von Projektdokumenten sowie die Definition eines neuartigen Benchmarks zur Messung narrativer Ähnlichkeit. Untersucht werden Prognosen von Erfolgsevaluierungen auf einer Skala von 1 bis 6, narrative Prognosen, Prognosen zur Kosteneffizienz sowie Vorhersagen binärer Outcome-Tags (wahr/falsch). Während keine der Methoden eine durchgehend hohe Genauigkeit bei der Vorhersage des Gesamterfolgs erreicht, zeigen statistische Modelle zuverlässig eine über dem Zufall liegende Fähigkeit, innerhalb von Maßnahmen derselben berichtenden Organisation und desselben Startjahres (sogenanntes „Pairwise Ranking“) erfolgreichere Maßnahmen zu identifizieren. Das vollständige Prognosesystem übertrifft bei der Evaluierung des Gesamterfolgs die Ergebnisse, die allein aus den in den Projektdokumenten angegebenen Risiken extrapoliert werden können. Dies zeigt sich in Tests mit einem zurückgehaltenen Testdatensatz von 200 der zuletzt gestarteten und evaluierten Maßnahmen aus einem Datensatz von 1181 umwelt- und nachhaltigkeitsbezogenen Maßnahmen, beschränkt auf vier berichtende Organisationen. Im Vergleich zur Baseline von 51 Pairwise Ranking für die Risikoextrapolation erreicht das ausgewählte statistische Modell unter Einbeziehung aller Eingangsmerkmale 60 [95%-Konfidenzintervall: 51 66] auf dem Testdatensatz. Über 14 binäre Outcome-Tags hinweg erzielen statistische Modelle ein durchschnittliches Pairwise Ranking von 60 (Spanne: 48% bis 77%) für die 45 der Paare mit unterschiedlichen Ergebnissen, gemittelt über zwei ausgewählte Methoden auf dem Testdatensatz. Diese Arbeit legt den Grundstein für eine verbesserte Entscheidungsfindung, wo immer umfangreiche Sammlungen von Beschreibungsdokumenten vor einer Intervention sowie zugehörige Evaluierungsdokumente nach Abschluss vorliegen

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1 Introduction

1.1 Research Questions

This thesis analyzes the applicability and composability of machine learning techniques in forecasting the success of international aid activities using activity information available at the approval stage. The standard approach in the literature for assessing aid impact and success in environmental domains has historically involved exclusively "statistical forecasting": econometric techniques such as fixed effects models, linear regression models, and occasionally modern nonlinear models such as Random Forest (RF) or XGBoost (Goldemberg, Jordan, and Kenyon, 2025) to statically analyze the relationships between features and overall success ratings. However, for statistical models to be most useful as tools to advise grant decision makers, they must generalize to unseen examples, in this case activities starting significantly after the start dates of activities in the IATI dataset. A thorough assessment of methods to forecast activity success in this domain is a core focus of this thesis.

In addition to statistical forecasting, this thesis proposes the use of Large Language Model (LLM)-implemented "judgemental forecasting" (Halawi et al., 2024). Judgemental forecasting allows forecasters to use tools including Fermi estimates, intuition, and information gathering to make calibrated predictions. It is particularly appropriate when explicit reasoning can improve outcomes beyond what purely statistical correlations reveal.

The modelling system developed in this work predicts how successful interventions will be in the context of international aid activities affecting the environment, utilizing statistical forecasting, judgemental forecasting, and a mixture of the two. Metadata and PDF files from thousands of activities were obtained, separating pre-approval information from post-evaluation outcomes. This data is used to answer the following research questions:

1. Can the overall rating, common true/false activity outcomes, or activity cost-effectiveness be forecast from pre-activity information alone?

2. How do judgemental forecasting methods compare to statistical models in forecasting international aid overall success ratings and common true/false outcome probabilities?

3. How do differing methods of combining language models and simpler statistical models compare in this domain?

4. What methods improve the accuracy of narrative (qualitative) forecasts in this domain?

5. What aspects of the activity available in the dataset at the beginning of the activity lead to higher or lower overall ratings, cost-effectiveness, or true/false outcome probabilities?

1.2 Evaluating Activity Forecasts

Activity outcomes are assessed using four separate evaluative approaches.

First, the overall evaluation rating, a numerical score assigned by the funding body at the time of evaluation, serves as the primary quantitative outcome. Second, narrative forecast similarity grades evaluate how closely an LLM-generated qualitative prediction matches the post-activity evaluation summary, providing a measure of forecast quality that is harder to game than ratings alone. Third, aggregate cost-effectiveness z-scores, derived from quantitative results extracted from evaluation documents, offer a proxy for the relative cost-effectiveness between activities. Fourth, in addition to quantitative cost-effectiveness outcomes and overall ratings, 14 separate true/false outcome tags pertaining to common activity outcomes were forecast.

Overall ratings are a standardized, commonly reported, and commonly assessed metric for international aid activities. A large literature examining patterns in overall ratings exists. However, because overall ratings determine future funding, they are subject to "gaming", where the rating is increased despite the activity not meriting that outcome (Goldemberg, Jordan, and Kenyon, 2025). Such gaming is inconsistent across activities and thus motivates forecasting other metrics of activity success.

Simply assessing whether a forecast of the overall rating is accurate may be misleading, as forecasts can be correct for the wrong reasons and unhelpful to aid decision makers. LLMs can output forecasts of how an activity will proceed in narrative form, which can then be qualitatively assessed against the narrative outcome summaries of evaluation documents. This provides an informative perspective on how different pre-activity information-gathering methods ultimately influence LLM forecasts. However, LLM forecasts are costly and companies that develop the best models are not transparent about their training process, harming reproducibility and introducing a greater risk of temporal leakage.

Another alternative to overall ratings is aggregate outcome cost-effectiveness. By extracting the quantitative achievements of activities and dividing by expenditure, one obtains a cost-effectiveness approximation that is roughly comparable across activities. This can be averaged with other indicators of cost-effectiveness such as reported benefit to cost ratio, or economic internal rates of return. Because quantitative achievements divided by expenditure are not often a key criterion for future funding, such a metric is less likely to be gamed. Cost-effectiveness is also less subjective than an evaluator's overall rating, and

is less susceptible to systematic differences between evaluators. While cost-effectiveness provides a useful alternative to ratings that are less likely to be gamed, accurate extraction of quantitative outcomes per activity and total expenditure is challenging. Challenges include lack of consistent reporting and definitions of quantitative units, differing options for which proxy for cost-effectiveness can be used, and differing definitions of total expenditure (such as whether expenditure is counted from only the reporting organization, or additional funding from recipient governments).

The fourth evaluative approach to assess overall activity success forecasts was a set of 14 extracted true/false outcome tags applied to each activity evaluation. Unlike overall rating and cost-effectiveness, the forecast is probabilistic. Outcome tags describe specific, discrete events or conditions that either occurred or did not occur in a project, such as whether the closing date was extended, whether targets were met, or whether funds were reallocated. Unlike the scalar overall success rating, outcome tags capture qualitatively distinct axes of project execution and make it possible to forecast specific types of project behavior rather than a single aggregate score. Thus, outcome tags provide a richer, more interpretable characterization of what happened in each activity, making the forecasting system substantially more useful as a decision support tool for grant officers than a single rating alone. Because several outcome tags were trained independently, claims about methodological improvements, feature usefulness, and feature importance ordering can all be stated much more confidently if they are consistent across several separate tags. Outcome tags objective, clearly comparable between activities, and do not suffer from differing definitions as much as the prior approaches. However, some outcome tags may not always be mentioned in outcome summaries.

1.3 Prior Work

An extensive literature review was conducted prior to constructing the forecasting systems. The key finding is that only (Goldemberg, Jordan, and Kenyon, 2025) is clearly comparable to this thesis, as it went beyond simply evaluating static patterns in overall activity ratings to testing generalization to unseen data via train/val/test splits, used nonlinear techniques rather than linear techniques as the rest of the literature does, and utilized LLMs and language model embeddings unlike other similar published work. Their work identified that "embeddings" and differences between embeddings of LLMs were the most important features used to predict country-level sector outcomes from the World Bank World Development Indicators (WDI).

One method that (Goldemberg, Jordan, and Kenyon, 2025) do not assess is forecasting, a core focus of this thesis absent from all aid evaluation literature known to the author. While many works in the literature have attempted to assess correlations between quantitative features and aid evaluation ratings, they have not focused on what knowledge

would be available at approval for the activity while simultaneously assessing out-of-time generalization of these correlations. Out-of-time generalization refers to model performance on activities that started after those used for training. Out-of-time generalization is critical: to provide advice on which proposed activity will be more or less likely to succeed, one must assess the ability of models to forecast the outcomes of currently planned aid activities, not simply statically analyze a corpus of past activities.

While the methodology appears novel within the domain of international aid evaluation, it has precedent in related academic domains.

Language Models for Forecasting Within the domain of LLM use, there have been early attempts at using LLMs to improve decision-making for the environment. A recent tool called "climsight" summarizes and aggregates information about climate adaptation and mitigation (Koldunov and Jung, 2024), but stops short of making forecasts for adaptation. Machine learning and LLMs have been used to collect over 80,000 articles about climate adaptation, identify gaps in implementation, and highlight underattended categories of policies (Callaghan et al., 2025).

Some work has also been published using LLMs such as GPT-4 to serve as data sources for policy deliberation and multi-criteria assessment of climate and sustainability interventions, finding that GPT-4 broadly agrees with the policy rankings of human experts (Bina et al., 2025). However, little was done to improve on GPT-4's abilities, the assessment covered only a few dozen generic policy examples, and no attempt was made to compare these policies against real-world outcomes. Despite these limitations, the findings are promising. For multiple criteria decision making (MCDM), GPT-4 provided a useful collaborative starting point, eased consideration of multiple criteria, and aided policy deliberation on climate change and sustainability.

In one similar work, Chain of Thought (CoT) reasoning was used to improve GPT-4's predictions of 1,261 conclusions from 276 papers analyzing field experiments in the social sciences. The study covers the impact of educational incentives, household finance behavior, healthcare enrollment, and financial planning. Over the 1261 outcomes, 78% were predicted accurately by the system (Chen, Hu, and Lu, 2025).

Another recent study found that LLMs can correctly predict results of published papers in neuroscience (Luo et al., 2025). This work used the raw probabilities generated by the language model rather than explicit reasoning, and for this reason was able to use much smaller language models than the GPT-4-scale models used in most other studies. Because language models work by assigning a probability of each token (typically some commonly occurring part of a word), multiplying the probabilities of all the tokens from the abstract allows researchers to compare the multiplied probability of the real abstract to the multiplied probability of the fabricated abstract directly, without having the language model generate any text involving reasoning or chain-of-thought.

Another study found a similar result with regard to publications in the domain of AI algorithms, finding their system beat human experts in predicting the ability of an AI algorithm to improve on the state of the art performance in AI models (Wen et al., 2025). In this domain, the researchers use a sophisticated framework with RAG and fine-tuning.

Another study used LLMs to predict the likely direction and effect size of empirical studies evaluating dietary interventions (Kaiser et al., 2025). This demonstrates that a fine-tuned LLM can predict direction of empirical intervention outcomes better than classical meta-regression baselines in dietary policy interventions, at an accuracy of approximately 80% to predict the directional outcome of the policy. The authors used a fine-tuned version of GPT-3.5 and carefully selected prompt variants that scored higher.

While the above studies demonstrate the promise of LLMs for forecasting intervention outcomes, they are largely domain-agnostic; the established literature on international aid evaluation has instead relied on classical statistical methods, with LLMs and embeddings only recently introduced by Goldemberg, Jordan, and Kenyon (2025).

Statistical Models in International Aid (Bulman, Kolkma, and Kraay, 2017) examined the micro and macro correlates of project outcomes across 3,797 World Bank and 1,322 Asian Development Bank projects using Ordinary Least Squares (OLS) linear regression, finding that project outcomes vary far more within countries than between them, with country-level variables accounting for only 10–25% of the variance. Among the significant predictors they identified were shorter planned project duration, the presence of additional financing, GDP growth, the country policy environment, and the track record of the project manager. (Ashton et al., 2023) extended this line of inquiry with additional World Bank data, confirming the importance of the country institutional setting and project supervision quality, and further identifying project design quality as a significant predictor of success; their OLS models explain up to 30% of the variance in project ratings on the training corpus.

(Eilers et al., 2025) analysed determinants of success and failure across 5,608 bilateral development finance project evaluations, finding that projects with more comprehensive financing are marginally more successful, that technical complexity and longer implementation duration are negatively correlated with success, and that higher risk is a significant predictor of failure. Their work draws on bilateral rather than multilateral donor data using the KfW Development Bank database, and introduces project-level risk and complexity as quantitative predictors, both of which are incorporated as features in the present work.

(Vivalt, 2020) investigated the generalizability of impact evaluations across contexts, finding that effect sizes from development interventions vary substantially between settings and implementing organizations, with government-implemented programmes tending to show smaller effects than NGO-implemented ones.

(Ndikumana and Pickbourn, 2017) used panel data methods to show that increased aid targeted to the water and sanitation sector in sub-Saharan Africa is associated with improved access to services, with governance indicators playing a moderating role. This underscores the relevance of country-level governance features for aid outcome prediction.

2 Methods

This section details how data were selected, filtered, and processed for each evaluative approach. Several methods to improve forecasting performance were selected based on validation set performance. A summary of the methods can be found at the end of this chapter in Section 2.13.

2.1 Data Selection and Sources

2.1.1 Unused Data Sources

The methods of this thesis required a suitable data source with all of the following: a sufficient volume of completed environmental and sustainability activities; paired pre- and post-activity documents to enable forecasting without future information leakage (see Section 2.2); machine-readable text extractable from database records; reliable temporal metadata; and the prospect of ongoing additions to allow future replication.

Several sources were evaluated against these criteria but ultimately not selected:

- **The OpenAlex publication repository of peer-reviewed evaluation documents and abstracts** (Priem, Piwowar, and Orr, 2022)
  - Although many millions of evaluations are available on OpenAlex, it proved difficult to reliably identify and deduplicate published evaluations of environmental interventions. Furthermore, OpenAlex provides only ex-post documents with no paired pre-activity description, making leakage prevention substantially harder.
- **The AidData database of international aid activities** (Tierney et al., 2011)
  - The AidData database uses double-coded data entry, which introduces less error than LLM and regex extraction techniques; however, the delay in hand-coding ratings and results yields relatively few recent environmental activities, and the absence of document links means each activity record contains far less extractable information. The overlap between AidData and the filtered IATI training set was less than 30% of IATI records. As AidData contained insufficient data for reliable forecasting on its own and overlap was low, merging or enriching IATI with AidData was not attempted.
- **The 3ie development database of ex-post evaluations** (*3ie Development Evidence Portal | 3ie* 2025)

  - 3ie had fewer than 300 records for environmental topics, far fewer than IATI, and provides only ex-post documents with no paired pre-activity description.

- **The IDEAL database of ex-post evaluations** (KfW Development Bank, 2025)
  - The IDEAL database, while better structured and organized than IATI, had fewer than 300 records for environmental topics. Some entries from BMZ/KfW/GIZ are also represented in IATI. IDEAL provides no pre-activity documents in its database.

2.1.2 Selected Data Sources

The IATI Dataset All activities for the thesis come from the IATI dataset (*IATI Dashboard – IATI Activities* 2025). IATI met all selection criteria: it contains a large number of activities with substantial textual information extractable from database records, and new records are continuously added, making it straightforward to add more activities for training or testing the model as they are added in the coming years. Crucially, IATI explicitly separates pre- and post-activity documents via a tag system, which makes preventing temporal information leakage easier (see Section 2.2). Records contain reliable "actual start" and "actual end" dates, and commonly also report planned start and end dates. Most IATI records contain an overall evaluation success rating on a six-point scale within the linked evaluation PDFs. Activity records also reliably mark the reporting organization. The records used for this thesis typically have several activity information documents uploaded near the beginning of the activity. Several years later, at least one final ex-post evaluation is uploaded on completion. The status of the activity is also always reported in the database, including if it is in the planning or completion/finalization stages.

A disadvantage of the IATI dataset is that it is sometimes inconsistent between reporting organizations. Additionally, documents were not always in PDF format, requiring conversion scripts from other formats. Many download links were also not functioning, or required custom web-scraping scripts to extract project documents in PDF format from the original hosting websites. Approximately 30% of IATI activities do not have an activity category code, leading to exclusion of environmental- or sustainability-related activities that would otherwise have been included.

Other Data Sources Country-level covariates were added from three standard sources: the World Bank governance indicators (Kaufmann and Kraay, 2024), the World Bank country policy and institutional assessment scores (World Bank, Operations Policy and Country Services, 2024), and GDP per capita (World Bank, 2024). These were included as supplementary features for the statistical forecasting models and were sometimes injected in the LLM forecasting prompts.

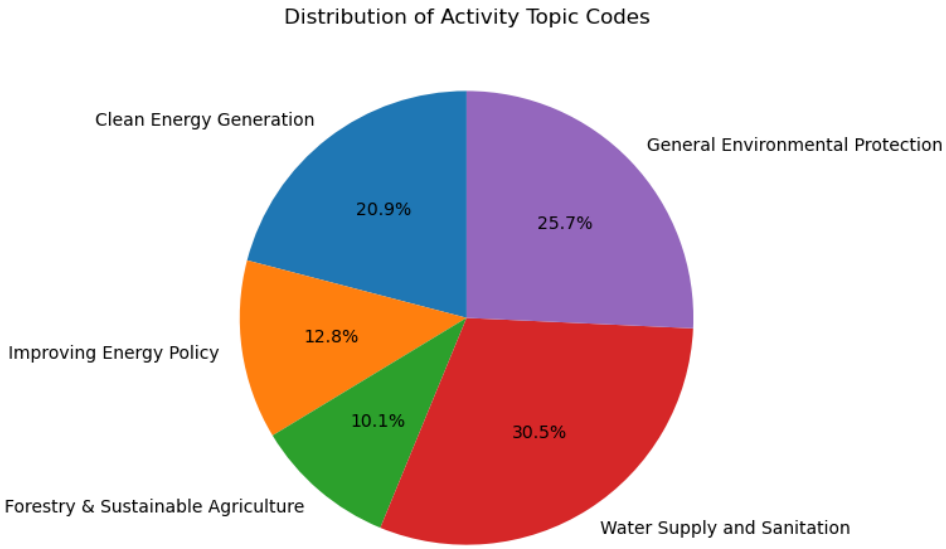

Figure 1: The split of topics analyzed from the dataset.

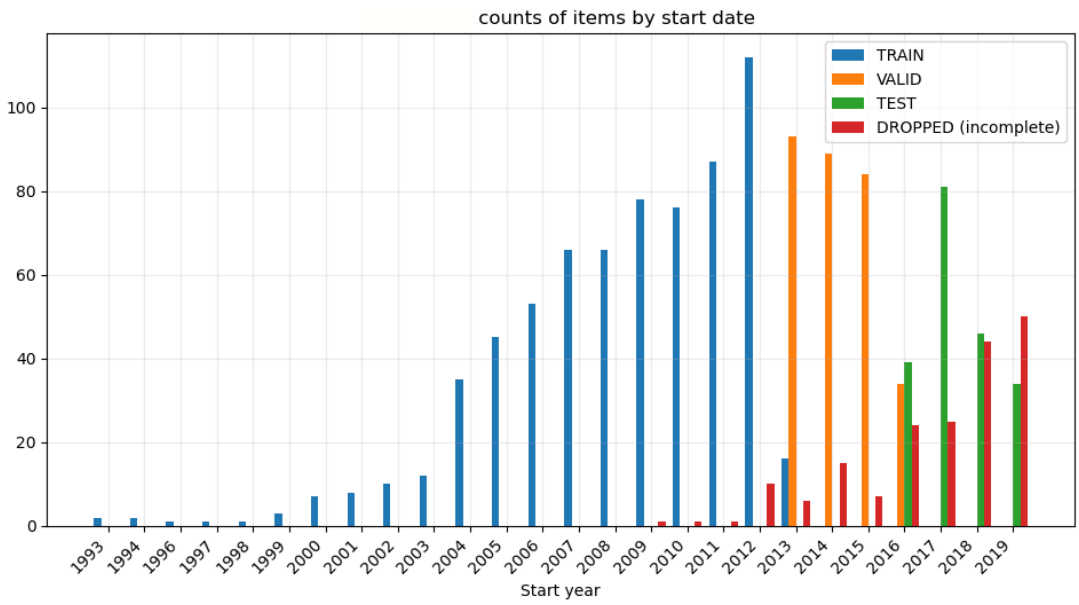

Figure 2: The activities included for forecasting ratings, with the splits by count year. Incomplete activities, shown in red, were not used for prediction. Activity ids used for outcome prediction were given the same temporal boundaries.

Table 1: Sizes of the four evaluation datasets used in this thesis, broken down by train/validation/test split. The AI forecasting set shows only the validation split (due to API cost and high rates of temporal leakage discovered on the test set).

Dataset	Train	Val	Test	Total
Overall ratings	682	299	200	1181
Outcome tags	681	299	196	1176
Cost-effectiveness	348	127	58	533
AI forecasting	–	283	–	283

2.2 Temporal Splits and Information Leakage

It is a common practice in developing statistical models with machine learning to train the model on a different set of data than one tests the model on. This motivates the separation of a "training" set from a "test" set of data. However, hyperparameters such as the number of trainable parameters in the underlying model are often selected empirically, and selection of the type of machine learning model is also frequently decided empirically. These decisions are made using a "validation" set of data which is distinct from the train and test sets, with the test set run only once to confirm that the choices of model and hyperparameters have a prediction skill similar to the validation set. As the focus of the thesis was forecasting, time-based splits were used, as is standard in statistical forecasting methodology. Train activities started on or before February 6, 2013; Validation activities started between February 7, 2013 and June 6, 2016; Test activities started after June 6, 2016 and before January 1, 2020. These date boundaries were chosen to yield approximately 700 training activities, 300 validation activities, and 200 test activities for overall ratings. Table 1 shows the resulting size of each split. See Figure 2 for the number of activities per year in each and Figure 4 for the average duration of activities within each split.

A forecasting system designed to predict the future must not use any information that could not have been known at the start of an international aid activity. The methodology was designed so that no information from the test set was used for model selection or validation, in order to realistically assess forecasting skill for activities starting in 2026 and beyond, consistent with BMZ grant officers' needs for a decision support system. Cases in which unavailable information was used for prediction are termed "leakage". Preventing leakage was a central methodological concern. Leakage has been significant in other published works on language model forecasting, and in multiple cases has invalidated claimed model skill in the geopolitical and economic forecasting domain (Paleka et al., 2025). While ex-post evaluations may adequately describe an activity, relying on language models to remove all references to eventual outcomes risks leakage of information from after the start date. This was a motivation for using IATI, which explicitly separates documents into pre- and post-activity, unlike OpenAlex and 3ie, which provide exclusively

ex-post evaluation documents.

Throughout testing, leakage incidents were detected and removed. For example, IATI reports planned start and end dates that are often retroactively altered to the true start date (verified by inspecting the PDF documents). The same is true for planned expenditure. Accordingly, planned duration and expenditure were extracted directly from pre-activity documents rather than from the reported IATI fields where possible. Similarly, the UMAP embeddings were at first constructed to include the test set, but were later restricted to only the validation set to ensure information about differences between embeddings could not inform the structure of the UMAP (see Section 2.6). In some cases pre-activity documents were in fact written later than the start date due to inaccurate metadata PDF dates, so pre-activity documents were screened by manual inspection, LLM-assisted checks, and automated keyword searches before being used for forecasting. Finally, a leakage script searching for common cases of leakage was run to detect and remove influenced LLM forecasts or LLM-generated features (see Section 3.1).

2.3 Language Model Choices

Several different language models are used for data extraction, the generation of semantic embeddings, LLM forecasting, and similarity grading, selected according to the following criteria:

Data Extraction LLM Model Selection

- *gemini-2.5-flash* was primarily used for its low cost, high speed, and ability to process PDF documents accurately. It was used for natural-language activity summaries, page summaries for pre-activity grading, page relevance grading, document ranking, quantitative outcome aggregation, finance breakdown and sector extraction, planned date and total expenditure extraction, and outcome tag discovery and taxonomy.
- *gemini-2.5-flash-lite* was used for high-volume requests where cost was otherwise prohibitive and task complexity was low, including page categorization and quantitative outcome extraction.
- *gpt-3.5-turbo* was used when the task benefited from an earlier model training cutoff date, eliminating the possibility of temporal information leakage for events occurring after September 2021. This legacy model is more expensive and has weaker reasoning ability than newer models. The data extraction use-case for *gpt-3.5-turbo* was to generate seven numerical feature grades based on the activity.
- *gemini-2.5-pro* was used when many pages of PDF context and extensive reasoning were required, with relatively few API calls. It was used to parse pre- and post-activity documents to generate mock forecasts (see Section 2.9.2).

Embedding Model Selection

- *gemini-embedding-001* embedded activity text for statistical model features and nearest-neighbour retrieval. It was selected over Sentence-BERT for its higher dimensionality and broader multilingual coverage.

LLM Forecasting Model Selection

- *deepseek-R1* was the primary forecasting model, selected for strong reasoning benchmark performance at low cost. It was used for both overall rating and outcome tag forecasting.
- *gemini-3-pro-preview* was used as a secondary forecasting model due to its superior reasoning benchmark performance, although it was used sparingly due to its high cost.
- *gemini-2.5-pro* was used for few-shot mock forecast generation from paired pre- and post-activity documents, selected for its ability to process long PDF contexts with high prompt adherence.
- *gemini-2.5-flash* was used for a fine-tuning experiment over 50 preference pairs for 20 epochs, selected as the only available model compatible with Vertex AI Direct Preference Optimization (DPO) fine-tuning.
- *gpt-3.5-turbo* and *gpt-4* were tested as forecasting models in a minimal prompt configuration without KNN or RAG context, serving as comparison baselines to assess the impact of model capability and training cutoff on forecasting skill.

Narrative Similarity Grading Model Selection

- *gemini-2.5-flash-lite* graded the similarity of LLM narrative forecasts to post-activity evaluation summaries, selected for its low cost on high-volume grading calls.

2.4 Data Filtering

The IATI dataset was progressively filtered from approximately 800,000 activities to 3,225 completed environment- and sustainability-related projects with both baseline and evaluation documents suitable for forecasting outcomes without future information leakage. Activities were first selected by sector codes (see Figure 1) and manual topic review. These were then constrained to those with downloadable documents and valid timestamps ("pre-activity" PDFs created no later than the end of the first quarter of the activity period as defined by the IATI start and end dates, "post-activity" PDFs created

no earlier than the final quarter). If a date was missing, standard titles such as 'Project Information Document' or 'Ex-Post Evaluation' were used to sort the documents into pre- and post-activity.

Additional restrictions ensured comparable rating scales and adequate sample sizes. The dataset was restricted to four reporting organizations with sufficient samples and compatible overall ratings: World Bank, BMZ/KfW/GIZ, ADB, and UK FCDO. See Figure 3 for the ratio of each organization per-split. These organizations were selected to ensure each had enough data for statistical models to learn per-organization differences.

After organizational filtering, documents were converted to PDF and dated using PDF metadata, which was validated against document content, yielding a median 22-day discrepancy. To prevent leakage from early progress reporting or misdated files, documents were screened by manual inspection, LLM-assisted checks, and automated keyword searches. Activities showing substantive outcome progress were excluded. Further filtering removed purely legal or contractual documents from baseline and evaluation documents. Activities not marked as completed in IATI were also removed, as were activities with no overall success evaluation ratings identified in the evaluation documents. This ultimately produced a set of 1181 activities used for the remaining analysis.

2.5 Data Processing

To accurately extract overall success ratings for each activity and useful textual information about the project for forecasting, each PDF document was processed using the following data processing pipeline:

- All document pages were rotation-corrected to vertical before processing via the Gemini API.
- Documents with ".odt", ".doc", or ".docx" extensions were converted to PDFs with a custom Python script.
- After conversion to PDF, zero- and one-page documents were excluded as insufficiently informative.
- Documents were then ranked and categorized, and structured information from the PDFs was systematically extracted via *gemini-2.5-flash*.

2.5.1 Ranking PDF Documents by Usefulness

Documents were first classified as pre-activity and post-activity (see Section 2.4). Pre-activity documents were created no later than $\frac{1}{4}$ of the way through the activity, and

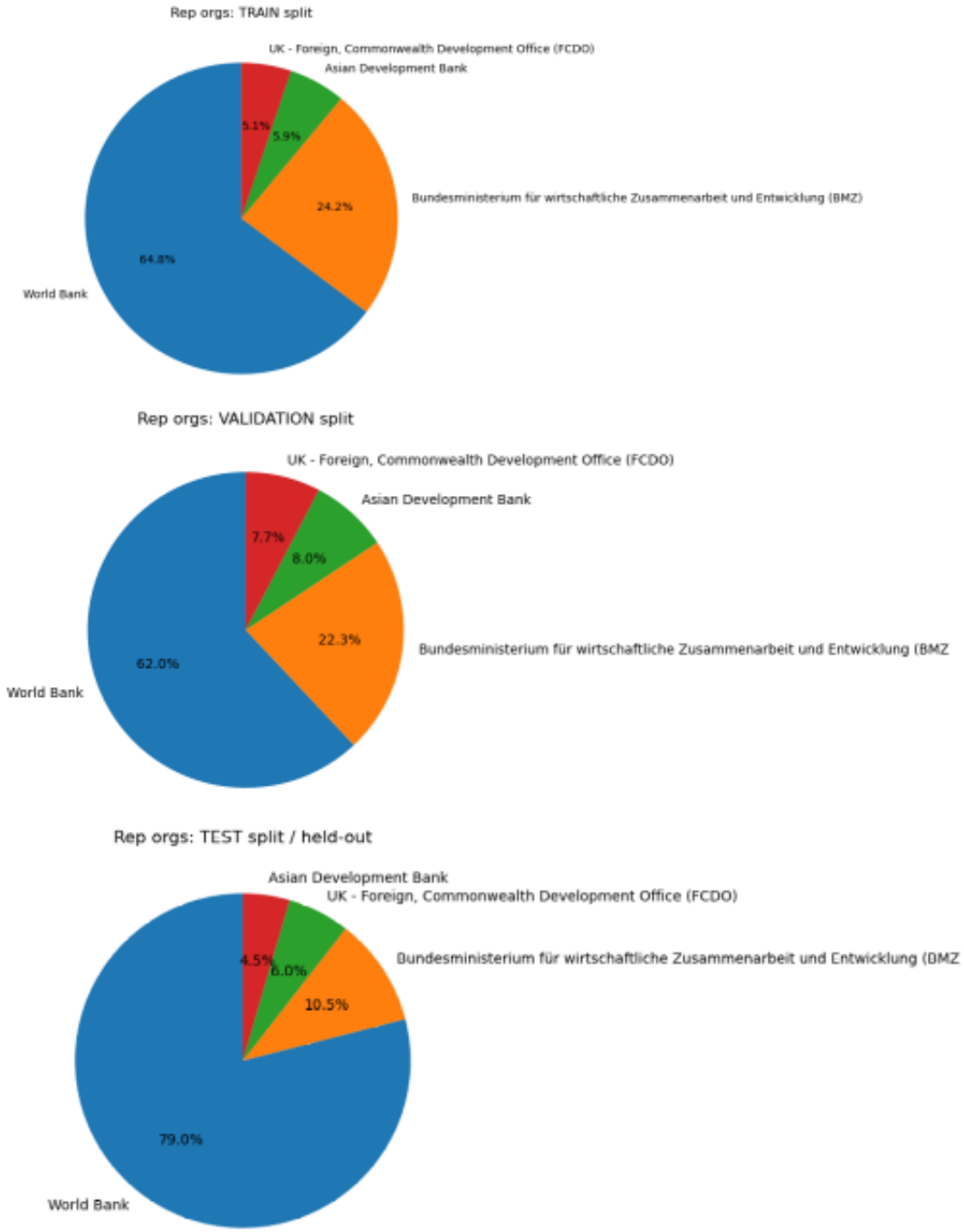

Figure 3: The breakdown of reporting orgs in the dataset which were used for training, validation and testing. The validation and test set contain differing ratios of reporting organizations between activities, making test set prediction more challenging.

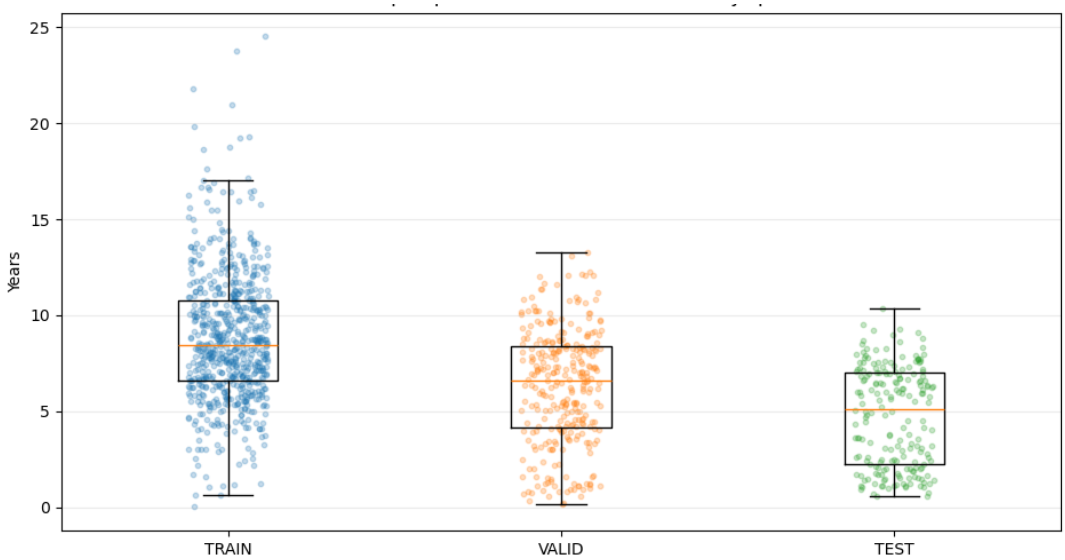

Figure 4: The durations of activities per split. More recently starting activities tend to be shorter, as they have not yet had time to complete and be evaluated. The out-of-distribution nature of validation and test sets increase the challenge of the model performing well on the test set.

post-activity documents no earlier than $\frac{3}{4}$ of the way through. Documents were then given a preliminary usefulness ranking without an LLM query. Pre-activity documents closest to the activity start and the most recent post-activity documents were prioritized, as were documents of 3–20 pages over very short or very long documents.

Documents were then given a final ranking with *gemini-2.5-flash*, ranking from most to least useful for either forecasting the outcome for pre-activity documents or evaluating the results for post-activity documents. Each activity was processed in two separate API calls: one *gemini-2.5-flash* query for pre-activity documents and one for post-activity documents. In each call, the first three pages of every candidate document were uploaded alongside a single prompt. This prompt contained the activity title, description, official date range, and per-document metadata (title, language, IATI document category codes, page count, and PDF metadata date). A rubric specifying the information an ideal document would contain was included to anchor the grading scale. The model was instructed to return a single structured JSON response ranking all candidate documents from most to least relevant, assigning each a usefulness letter grade (A+ to F) and an include/exclude flag. Letter grades followed standard grading examples from Google Gemini documentation. Structured output was enforced via a JSON schema so that every document received exactly one rank and grade in a single response. Non-English duplicates were excluded when an English equivalent was available. When multiple progress reports existed, all but the latest were excluded from the rankings. After ranking, 2,312 documents had sufficiently informative activity information and activity evaluation documents.

2.5.2 Categorizing Pages within PDF Documents

Each page of the highest-ranking documents was then categorized by its content. Comparison with a small set of activities using standard RAG retrieval showed that this approach was more effective than semantic embedding search for long documents. PDFs were split into 3-page chunks, each sent to *gemini-2.5-flash*. Several documents were read and categorized by hand to identify common sections, which informed the category set used for automated categorization. The pages were categorized differently based on whether the document was a pre-activity or post-activity document. Categories for post-activity PDF pages broadly covered whether the page contained quantitative or qualitative final evaluations, deviations from plans, other post-activity content, or merely an overview of the activity. Specifically, the allowed categorizations for pre-activity document pages were "condensed summary", "sub activities outlined", "detailed implementation plans", "broad objectives", "possible outcomes", "quantitative targets", "qualitative targets", "risks as word or numeric", "risks or dangers generally", "plans to address key risks", "positive indicators", "progress reports", "similar cases outcomes", "implementation context country", "contextual challenges", "financing details", "budget and legal", "who implements", "whether part of larger program", "partner identity or skill", "whether skin in the game", "other stakeholder engagement", or "activity monitoring details". For post-activity document pages, the allowed categorizations were "expected outcomes", "deviation from plans", "preliminary results", "final outcomes", "delays or early completion", "over or under spending", "overview as was planned", or "unrelated to evaluation".

Attempting this categorization scheme by hand during development revealed that pages often pertained to more than one category such as a page which documents final outcomes while also documenting project delays. The two most relevant category choices among these were therefore assigned for each page in the documents. Only two categories were allowed, to prevent less relevant categories from polluting the page tags.

To exclude irrelevant pages, the pages were also given a second categorization type, for post-activity document pages as "glossary", "blank page", "table of contents", "outcome evaluation", "activity description", "references", or "other", and for pre-activity document pages the same categories were options, in addition to "core activities", "theory of change", "targets", "broader context", and "preliminary results". Only one category was assigned per page, as these categories are typically mutually exclusive.

2.5.3 Post-Activity Structured Information Extraction

2.5.3.1 Extracting Overall Activity Success Ratings

Two separate methods were used to extract overall activity success ratings. Inspection of randomly selected pages revealed that ratings often appeared on pages with a high

usefulness letter grade, with a threshold of approximately 7/10 separating matches from poor fits. The first method therefore sent each post-activity document page rated above 7/10 for relevance to evaluation, or categorized as "quantitative targets", to *gemini-2.5-flash* to extract any overall ratings; a second script then consolidated these into a single value per document. However, this was often insufficient to capture the overall ratings. A fallback script used a custom word search of approximately 500 rephrasings of "overall rating", "final result", "synthesized score", and similar terms in English, searching PDFs directly for exact matches and prioritizing pages with one or more such matches. If no matching terms were found, the earliest non-excluded pages (those not categorized as "blank page", "appendix", "glossary", "table of contents", "references", or "activity description") were concatenated into a single PDF per activity, and *gemini-2.5-flash* was queried to extract the overall rating.

For BMZ/KfW/GIZ activities, pre-activity documents were extremely rare; accordingly, the evaluation document was treated as a pre-activity document for the purposes of forecasting activity success. Categorization for these evaluations used the pre-activity document method described above. When grading or summarizing the evaluation document, *gemini-2.5-flash* was instructed to describe only what could have been known at the beginning of the activity and under no circumstances to reveal the final outcome. Manual inspection of a random sample of extracted information from these documents did not reveal any resulting temporal leakage. BMZ/KfW/GIZ overall ratings were similarly predictable to World Bank ratings on the test set despite this leakage risk. BMZ/KfW/GIZ activities comprised approximately 16% of the overall dataset, with around 17% of activities in the training and validation sets but only 10% in the test set.

Ratings were reported both with the rating itself, and a maximum and minimum possible rating to establish the scale. The World Bank rating scale from 1 ("Highly Unsatisfactory") to 6 ("Highly Satisfactory") was used as the template rating, and other rating systems were rescaled to match the 1 to 6 rating scale. BMZ/KfW/GIZ ratings were inverted to match this scale. A "Satisfactory" score was considered equivalent to scores such as "successful", "On Track", or "met expectations". Scores listed as percentages or fractions were re-scaled to the 1 to 6 scale as well.

2.5.3.2 Extracting Cost-Effectiveness

A similar approach using *gemini-2.5-flash* and *gemini-2.5-flash-lite* was applied to extract quantitative cost-effectiveness-related outcomes.

Cost-effectiveness of international aid activities was defined as the ratio of measured quantitative outcome aiming to improve the environment or sustainability to the expenditures used to achieve that outcome. Quantitative outcomes were extracted by language model prompts as described below (the numerator). The majority of international aid activities in IATI directly publish the overall activity expenditure (the denominator).

Quantitative outcomes were extracted from post-activity documents already marked as containing quantitative outcomes by the categorization script. A structured JSON response was saved with the description of the outcome and the outcome value itself, along with the units of the quantity. A combination of manual examination of the extracted outcomes and a frequency analysis of bigrams (pairs of words) was used to identify outcome variables that could be compared between projects. For each common outcome category, a list of words and phrases was compiled that would commonly match reports of these outcome variables in the description, as well as appropriate units. Each script ensures implausible or non-numeric values are dropped. Domain-specific filters also reduced false positives (e.g., wastewater context for pollution loads, water context for connections, agriculture context for yields).

Comparable Outcome Categories Using manual inspection and frequency statistics of common bigrams, a set of outcome categories was defined that recur across evaluations and are interpretable across projects. One challenge was ensuring sufficient quantities of comparable outcomes between activities. Only outcome categories containing at least 50 activities in the training set were retained. This threshold was selected to ensure reliable validation performance: below 50 activities model performance declined, while a higher threshold left too few points to evaluate the final prediction within a reasonably narrow confidence interval. The model was trained and tested on the following outcome categories:

- **Benefit/cost ratios (B/C):** benefit-cost ratio outcomes. (train: 62, val: 18, test: 15, total: 77)
- **Rates of return:** economic rate of return (ERR/EIRR) and financial rate of return (FRR/FIRR), in percent. (ERR — train: 396, val: 103, test: 43, total: 440; FRR — train: 190, val: 43, test: 15, total: 205)
- **Emissions reductions:** CO_2 or CO_2e reductions in tonnes. (train: 61, val: 23, test: 10, total: 72)
- **Water and sanitation connections:** counts of service connections, either new or repaired. (train: 96, val: 25, test: 9, total: 105)
- **Energy outcomes:** installed generation capacity, in MW (or occasionally GWh where the source reported capacity in energy units). (train: 64, val: 20, test: 12, total: 79)

Although pollution load removed, forest indicators, trees planted, air quality (PM2.5), clean cooking stove distribution, and agricultural yield improvement were parsed, none contributed sufficient counts to be useful for further outcome forecasting.

All extracted quantitative outcomes within the same activity and outcome category were aggregated using *gemini-2.5-flash* to produce a final baseline, target, and outcome value. Where multiple extracted values within the same activity and outcome category

conflicted, overlapped, or reflected different levels of aggregation (e.g., component-level versus project-level figures), the model was instructed to adjudicate between them and select or combine values to reflect a single coherent project-level outcome. The model was instructed to distinguish duplicates from additive components, prioritize project-level totals over subcomponent figures, aggregate only non-overlapping quantities, and return a null response when no coherent representative value could be determined.

Quantitative Outcome Aggregation and Cost-Effectiveness Computation Once outcomes were extracted and aggregated, cost-per-unit estimates were constructed. The total expenditure converted to USD for each activity reported by IATI served as the cost denominator, with the exception of benefit-cost ratios and rates of return, which are inherently unit-free or already normalised.

Outcome-level funding splits are not published in the IATI dataset. To avoid implicitly treating multi-outcome activities as having multiple full budgets, a custom algorithm allocates total activity expenditure across the non-overlapping funded sub-activities and activity reports. Benefit/cost ratios and economic and financial rates of return are excluded from monetary allocation; all other outcome categories are eligible including those with fewer than 50 training set activities. To avoid double-counting when two indicators are alternative measurements of the same underlying result, closely related indicators are first grouped into shared conceptual buckets (e.g., protected area paired with area under management; different yield-increase measures; tree planting paired with reforested area).

Once the components are bucketed, each bucket receives an equal share of the activity's allocatable funding. Every component inside a bucket inherits that same share, so "alternative measures of the same thing" draw on one portion of the allocation rather than each taking a separate slice.

Carbon dioxide reductions are handled as a special case because they can act as a summary metric that overlaps with other mitigation outputs. If CO_2 reductions are reported without any closely linked mitigation outputs (such as improved stoves, added generation capacity, or trees planted), CO_2 reductions receive an equal share like any other bucket. If CO_2 reductions are reported alongside any of those linked outputs, they inherit the combined allocation already assigned to the CO_2-mitigating expenditures for that activity, preventing CO_2 from inflating allocated spending when it is a co-reported consequence of other outcomes.

Most cost-per-unit outcomes are approximately log-normally distributed. The $\log_{10}$ transform was therefore applied to all outcome categories except benefit-cost ratios and rates of return before modelling.

2.5.3.3 Extracting Outcome Tags

All evaluations were summarized, common outcome tags were identified from a random

sample of these summaries, and the value of each true/false tag was then determined for all outcome summaries.

Extracting Outcome Summaries For each activity, up to 10 post-activity document pages were selected for outcome summarization, prioritising pages categorised as containing deviations from plans, delays or early completion, or over- or under-spending. Surrounding pages were included when insufficient high-relevance pages were available. Manual inspection revealed that below a threshold of 3/10 relevance, pages were typically not relevant to reasons for ratings or explaining outcomes. A minimum relevance score of 3/10 was therefore applied. If an activity had no categorised pages meeting the threshold of 3/10, a fallback script retrieved the highest-scoring uncategorised post-activity pages directly from the document. These pages, together with the activity title, planned date range, scope, and the activity description from IATI, were submitted in a single *gemini-2.5-flash* prompt. The model was instructed to return a bullet-point summary of what occurred as a result of the activity, reporting both successes and shortfalls and any quantitative outcomes relating to the activity's core aims while explicitly withholding the overall evaluation rating and avoiding introductions or conclusions. The resulting post-activity evaluation summaries were used as the input for outcome tag classification.

Discovery of Common Tags To determine which tags should be applied, a random sample of 250 evaluation summaries from completed activities was drawn. This sample was divided into batches of 10 evaluation summaries each. Each batch was submitted to *gemini-2.5-flash* with a prompt requesting identification of all recurring outcome types present across the summaries in the batch. A set of seed examples determined from manual inspection of random activity summaries was provided to anchor the initial vocabulary (e.g., "implementation delays," "high beneficiary satisfaction," "project was restructured," "capacity building training was delivered"). The model was instructed to generate additional tags for any distinct outcome type appearing in more than one summary. All 25 batches were then deduplicated and merged via manual inspection, resulting in 23 outcome tags covering finance and budget outcomes, activity rescoping outcomes, process and implementation challenges, and target achievement outcomes.

Unsigned and Signed Tags Of the 23 tags, 11 were treated as false (not having occurred) if not mentioned; these are referred to as "signed" tags. They cover phenomena applicable to any project that would very likely be mentioned in a summary if they occurred, such as whether the project was delayed. The remaining 12 tags were recorded as null (neither true nor false) if not mentioned; these are referred to as "unsigned" tags. They were recorded only when explicitly mentioned in the summary, such as whether infrastructure was successfully completed or livelihoods were significantly improved. For unsigned tags, the LLM is asked whether the phenomenon was clearly described as occurring, and the response is a single boolean per tag. See Section 2.11.2 for the chosen tags and whether they were signed or unsigned.

Applying Tags at Scale The finalized set of tags was extracted for all activities with overall activity ratings within the four reporting organizations using two separate LLM passes over each activity's post-activity outcome summary. In the first pass, all 12 unsigned tags were evaluated together in a single prompt per activity, with the model returning a JSON object enforced via a structured output schema. In the second pass, the remaining signed tags were extracted similarly.

2.6 Input Features for Forecasting Models

All methods used input features to improve forecasts. The generation method for each feature is described below.

Non-LLM Features Sources are cited below which have found meaningful predictive effects or correlations between activity success or activity outcomes and the features listed.

The non-LLM features used were:

- planned activity duration (Goldemberg, Jordan, and Kenyon, 2025)
- planned total disbursement (consultation with BMZ officers), (Bulman, Kolkma, and Kraay, 2017) (Goldemberg, Jordan, and Kenyon, 2025) (Ashton et al., 2023) (Eilers et al., 2025)
- derived features: log(planned total disbursement), planned total disbursement per year
- whether the activity is primarily loan or grant-based (Bulman, Kolkma, and Kraay, 2017)
- the one-hot encoded reporting organization (Goldemberg, Jordan, and Kenyon, 2025)
- the Country Policy and Institutional Assessment (CPIA) score from the World Bank for that country (Bulman, Kolkma, and Kraay, 2017)
- the mean of 5 World Governance Indicator scores (Kaufmann and Kraay, 2024). The mean was taken of control of corruption, government effectiveness, political stability, regulatory quality, and rule of law (Goldemberg, Jordan, and Kenyon, 2025) (Ndikumana and Pickbourn, 2017)
- the scope of the activity on a scale from 1-7, ranging from local to global (Vivalt, 2020)
- the log(GDP/capita) of the countries where the activity takes place weighted by the percentage of the activity performed in each country. (Goldemberg, Jordan, and Kenyon, 2025) (Bulman, Kolkma, and Kraay, 2017)
- A dummy variable for each region as defined by the World Bank (AFE, Eastern and Southern Africa; AFW, Western and Central Africa; EAP, East Asia and Pacific; ECA, Europe and Central Asia; LAC, Latin America and the Caribbean; MENA,

Middle East and North Africa; SAS, South Asia) (Goldemberg, Jordan, and Kenyon, 2025) (Vivalt, 2020)
- missingness counts for important features with missingness

The activity start date was not used as a feature, as there was no linear pattern with regards to overall activity success over time in the training data, and adding it tended to increase overfitting and reduce out-of-time generalization. As opposed to some results in prior literature (Vivalt, 2020) (Ashton et al., 2023), whether the implementer was a government, NGO, or another category was also not found to have any improvement in model performance on the validation set and was excluded.

LLM-generated Grades Seven additional features were extracted using *gpt-3.5-turbo*-generated grades to enrich the activity with information from the pre-activity document PDFs. *gemini-2.5-flash* first created a summary of the relevant information from the document, and *gpt-3.5-turbo* was used to provide a final grade on a scale from 0 to 100. *gpt-3.5-turbo* was selected to reduce leakage of information after the September 2021 model cutoff date. Inspection of *gemini-2.5-flash* summaries revealed temporal information leakage occurred in grades for 3% of the train+val activities, in all cases due to COVID being mentioned despite the activity start date occurring before the 2019 onset of the pandemic (see Section 2.2). Grades from summaries with leakage were removed and replaced with the train+val median grade. The majority of activities in the test set ended after the model cutoff date. These seven features were based on important aspects revealed by the literature review and consultation. One feature not introduced by others was added (the ease of targeted outcomes). Unlike technical complexity, which captures only one dimension of difficulty, or overall risk, which aggregates financial, legal, and regulatory threat sources, ease of targeted outcomes is distinct: it specifically reflects whether the activity's intended outputs are inherently measurable and achievable given typical implementation constraints, a factor theorized to independently contribute to evaluation success.

The features added for this method were the seven grades and one interaction term:

- *gpt-3.5-turbo*-generated grade from 0 to 100 of:
  - how well financed the activity is (consultation with BMZ officers), (Bulman, Kolkma, and Kraay, 2017) (Goldemberg, Jordan, and Kenyon, 2025) (Eilers et al., 2025)
  - the activity integratedness within the broader activity ecosystem (consultation with BMZ officers), (Vivalt, 2020)
  - the expected implementer performance including ownership (consultation with BMZ officers), (Bulman, Kolkma, and Kraay, 2017) (Goldemberg, Jordan, and Kenyon, 2025) (Ashton et al., 2023)
  - the degree of contextual challenge (consultation with BMZ officers)

  - the overall risk level (consultation with BMZ officers), (Eilers et al., 2025)
  - the activity's overall technical complexity (Eilers et al., 2025)
  - the ease of targeted outcomes (not found in literature, but added to assess how measurable and achievable the activity goals are)

- Additional interaction term added

  - expenditure times complexity

LLM-Generated Budget Allocation for Sector Clustering Budget allocations are an important factor for activity success (Goldemberg, Jordan, and Kenyon, 2025). *gemini-2.5-flash* was queried to identify the breakdown of outcome-related funding, where available in the pre-activity documents. Unlike in (Goldemberg, Jordan, and Kenyon, 2025), the measure of how much the funding allocations are split between different domains (the Herfindahl–Hirschman index (HHI)) was found not to improve performance on the validation set and was not added as a feature. Possible reasons for this include that the environmental activity subset may exhibit less variance in funding concentration than the full IATI portfolio, that sector clustering already partially captures the dispersion signal HHI summarises, and that HHI may be less discriminating within a domain where activities are already thematically concentrated. The allocations of funding are required to approximately sum to the total expenditure of the project, with 68% of projects having their budgets identified. If no high-scoring finance or budget pages were identified in the categorization step, the first 10 pages of the highest-ranked pre-activity document were used as a fallback.

To make the resulting dataset of budgets usable by the statistical model, the embeddings model was used to cluster the descriptions of the subsectors into 15 sector clusters. Having tried 10, 15, and 20 clusters, 15 was found to provide the optimal performance on the validation set. The allocations within each cluster are summed and reported as a fraction of the total as a feature for the statistical model.

This adds the following 16 features to the statistical model:

- 15 features, one for each sector cluster expenditure percentage (Goldemberg, Jordan, and Kenyon, 2025)

- a binary missing indicator if there is no sector cluster information in the PDF. As for other features, the result is median imputed, meaning all features are set to 0 as that is the median for all sectors in the training data.

Language Model Embeddings The underlying information contained in text can be mapped onto a high-dimensional vector of numbers. This process is called "embedding." The textual input is the LLM-generated "targets" field, which summarizes the activity's

objectives. This follows (Goldemberg, Jordan, and Kenyon, 2025) in extracting the World Bank Project Development Objectives (PDO), as they are found to be very similar to the LLM-generated outputs. This was verified by manual inspection of approximately 50 random samples, comparing outcome text generated by the model with the text from the PDO document; both tend to cover similar content and format. PDO extraction was also attempted using regex methods, but the results were found to be noisy on matching, especially on projects not from the World Bank, and did not improve prediction performance as much as embeddings on the LLM-generated targets. The LLM-extracted targets text was first normalized into a stable canonical form (removing formatting artifacts, unescaping, splitting on separators, dropping "NO RESPONSE" tails, and deduplicating near-identical chunks). The cleaned targets text for each activity was then embedded with the *gemini-embedding-001* model, yielding a single high-dimensional vector per activity that semantically encodes activity objectives.

Following (Goldemberg, Jordan, and Kenyon, 2025), target embeddings are compressed using a two-stage dimensionality reduction pipeline. This works via Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) (Healy and McInnes, 2024). PCA projects embeddings into a few directions that preserve most of their variation by finding orthogonal directions along which the data varies most, then projecting the data onto those directions. UMAP builds a graph of nearest neighbors in high-dimensional space, then optimizes a low-dimensional embedding to preserve those relationships. PCA was first used to reduce the embedding vectors to 50 dimensions and then UMAP was fitted on the PCA outputs to produce 2D and 3D coordinate maps. Applying PCA before UMAP in this way is a standard practice recommended in (Healy and McInnes, 2024): it removes dimensions that are dominated by random noise while retaining the principal axes of variation, so that UMAP's nearest-neighbour graph is built on a cleaner signal and produces more stable projections. 3D embeddings (umap_x, umap_y, umap_z) outperformed 2D and 4D on the validation set for forecasting activity ratings, so 3 dimensions were used. Because the evaluation uses an out-of-time split, PCA and UMAP are fitted only on activities starting before the relevant evaluation cutoff: training-period activities only when evaluating on the validation set, and train+validation activities when evaluating on the test set. Test activities are never included in the manifold fit. Instead, their PCA-compressed embeddings are projected into the pre-existing UMAP space. This prevents any future-activity information from influencing the coordinate system, at the cost of somewhat noisier projections for activities lying outside the training distribution. The same cutoff logic applies to the sector and country centroids used for the distance features described below. Visual inspection also revealed that sectors with similar 2D vectors cluster around the activity environmental category, replicating the finding in (Goldemberg, Jordan, and Kenyon, 2025).

Word-frequency comparisons across the UMAP x, y, and z axes reveal what aspects the

embeddings capture and the sectors in which activities are categorized. Broadly, low x values correspond to forestry, agriculture, water, and management-related terms, while high x values correspond to energy-sector and financing-related terms. Low y values correspond to similar terms as high x, mostly in the energy sector, while high y values relate to biodiversity, wastewater, and conservation. The z axis appears to reflect an urban–rural dimension: low z corresponds to wildlife and UNDP-related terms and "rural," while high z corresponds to sanitation, wastewater, "urban," and "city." The z axis may also correlate weakly with the typicality of objectives within a sector, with more generic objectives associated with lower z values. Finally, the z axis shows a weak negative correlation with the number of countries, with lower z values tending to occur in activities with a smaller geographic scope.

This method adds 4 additional features to the model:

- umap3_x: First UMAP coordinate capturing a spectrum from forestry, agriculture, and water management objectives (low values) to energy-sector and financing-oriented objectives (high values).
- umap3_y: Second UMAP coordinate separating energy-focused activities (low values) from biodiversity, wastewater, and conservation-oriented activities (high values).
- umap3_z: Third UMAP coordinate primarily reflecting an urban–rural axis, with lower values associated with rural and wildlife objectives and the UNDP, and higher values associated with urban sanitation and wastewater.
- a binary indicator for whether the umap embedding is missing (missing features are always imputed using the median values in the training dataset)

Language Model Embedding Contextualization Beyond the compressed UMAP coordinates, two scalar distance features are derived from the full-dimensional target embeddings. For each activity, sector distance is the Euclidean distance between the activity's target embedding and the centroid of all target embeddings belonging to the same sector and decade group. Country distance is the distance between the activity's embedding and the weighted sum of the recipient countries' embedding centroids, where the weights are the fraction of the activity carried out in each country, again grouped by decade. If no country data are available, the decade-level centroid is used as a fallback. Both were found to be informative on the validation set and are included in the full feature set.

This approach adapts and extends the methodology of (Goldemberg, Jordan, and Kenyon, 2025), who use Sentence-BERT embeddings of Project Development Objectives and project titles to construct sector- and country-level distance measures for World Bank projects,

and show these embeddings correlate with quantitative country-level sector outcomes. The key differences here are that the present work uses *gemini-embedding-001* rather than Sentence-BERT, trains on LLM-generated "targets" text rather than raw PDO strings (enabling coverage beyond World Bank activities), and computes distances at the activity level for individual success prediction rather than in aggregate for country-level outcome analysis.

This method adds 2 additional features to the model:

- sector_distance: Euclidean distance between the activity's target embedding and the centroid of target embeddings for activities in the same sector and decade. Low values indicate the activity's objectives are typical for its sector; high values indicate an unusual focus.
- country_distance: Euclidean distance between the activity's target embedding and the weighted sum of per-country centroids for the recipient countries, weighted by the fraction of the activity carried out in each country. Low values indicate the activity's objectives closely resemble the typical portfolio of its recipient countries; high values indicate objectives that are atypical for those country contexts.

2.7 Shared Experimental Methods

The Random Forest and Extra Trees ensemble (RF+ET) and start-year trend correction were applied across all activity success forecast methods, excluding the LLM-only forecasts.

Random Forest The Random Forest (RF) is a statistical algorithm that constructs an ensemble of decision trees fitted to the training data and averages their outputs, reducing overfitting. The decision trees split based on value ranges of the features. By reducing tree depth (the number of split points), memorization of the training data is reduced and generalization improves. Each split also considers only a random fraction of features, increasing tree independence and improving generalization. The bootstrap method was also used to train trees, encouraging tree independence.

Extra Trees The Extra Trees (ET) method, short for Extremely Randomized Trees, is a variant of the RF algorithm that constructs an ensemble of decision trees with an additional layer of randomization. While RF selects the optimal split point for each randomly chosen feature subset, ET goes further by also randomizing the split thresholds, drawing cut points uniformly at random from each feature's value range and selecting the best among those random candidates. This additional randomization further decorrelates the individual trees in the ensemble, reducing overfitting at the cost of a slightly less precise fit to the data. Unlike RF, ET trains each tree on the full training dataset rather than a bootstrap sample. This is beneficial when ensembling with RF, as the two models

then derive their diversity from different sources: one from varying the data each tree sees (bootstrap), and the other from randomizing split logic (ET). This makes their errors more complementary and the ensemble average more accurate. As with the RF, predictions are made by averaging the outputs of all trees in the ensemble.

Tree Parameters Adjustments to RF+ET parameters were found to strongly influence model performance. Optimal parameters for the overall rating RF+ET were discovered via a combination of sweeps over parameter space and LLM-generated suggestions iterating on the identified best parameters. The default `RandomForestRegressor` hyperparameter configuration uses `n_estimators`=100, `max_depth`=`None`, `min_samples_split`=2, `min_samples_leaf`=1, `max_features`=1.0, `bootstrap`=`True`, and `ccp_alpha`=0.0. The best-identified specification, with `n_estimators`=638, deliberately constrains model capacity in ways that typically reduce overfitting. Trees are explicitly depth-limited (`max_depth`=14 rather than unbounded) and splits are only permitted when nodes contain substantially more data (`min_samples_split`=20 and `min_samples_leaf`=20), which smooths forecasts by limiting fine-grained partitioning of the feature space. In addition, using a smaller feature subset at each split (`max_features`=0.488) increases tree diversity and reduces variance relative to the default that considers all features. The model also uses row subsampling (`max_samples`=0.86), further reducing variance by injecting additional randomness into each tree's training set. Compared to defaults, these choices accept slightly less precise fits on the training data in exchange for predictions that are more stable across unseen examples.

The same parameters as overall ratings were used for the cost-effectiveness model.

The outcome tags used both a uniform set of tree parameters and a per-tag set. Details for the tree parameter selection for outcome tags can be found in Section 2.11.2.

Start-Year Trend Correction After training, all RF+ET predictions were corrected for a linear temporal trend in the residuals of the RF model. For all RF+ET models, a ridge regression (penalty $\alpha = 50$) was fitted on training data to predict the residual $r_i = y_i - \hat{y}_i$ from the activity start year:

$$\hat{r}_i := \gamma_0 + \gamma_1 \cdot \text{year}_i,$$

and the corrected probability was:

$$\hat{y}_i^{\text{corr}} := \text{clip}(\hat{y}_i + \hat{r}_i)\,.$$

where the clip range is [1,6] for overall ratings and [0,1] for outcome tag probabilities. The strong ridge penalty shrinks the slope toward zero to avoid overcorrecting on noisy year–residual relationships. The correction multiplier was validated by sweeping values in $\{0, 0.1, 0.2, 0.5, 1.0, 1.5, 2.0\}$ and evaluating mean within-group pairwise ranking, Brier

skill, and accuracy skill on the validation set across outcome tags. A multiplier of 1.0 (full correction) achieved the best mean pairwise ranking, Brier skill, and accuracy on the outcome tags, and was therefore retained for all year trend corrections. The correction was applied to the RF model alone before averaging with ET, as this was found to outperform correcting the ensemble average on the validation set. The correction was applied to overall rating prediction, cost-effectiveness, and all tags except the monitoring and evaluation tag and the high disbursement tag, where the year trend was found to hurt validation set performance. Year correlation effectiveness was additionally separately assessed on the test set for overall ratings as an experimental method.

2.8 Forecasting Overall Ratings with Statistical Models

2.8.1 Overall Rating Baseline Methods

Always Predict the Most Common Rating for the Reporting Organization This baseline method provides a sanity check that more sophisticated methods are worthwhile. If this baseline outperforms more sophisticated models, it indicates that the methodology is overfitting on noise in the data.

Ridge Regression with Reporting Organization and Risk Score To assess how well pre-activity risk assessments are calibrated, a ridge regression model was trained using the LLM risk grade and a dummy variable for the reporting organization. Although the full forecasting system also includes these features and should therefore outperform this baseline, it provides insight into how much of the eventual rating can be inferred from pre-activity risk assessments alone. This score was only calculated for activities where the LLM was able to successfully extract the grade for how risky the project is from the pre-activity documents. This method suffers from potential inconsistency in LLMs extracting the overall risk from pre-activity documents.

RF+ET Model Trained with Non-LLM Categories To justify the addition of non-LLM input features, the model is restricted to features not generated or extracted by LLMs. Prediction skill can then be compared to the full model to verify that LLM-extracted features improve performance.

2.8.2 Overall Rating Experimental Methods

This thesis introduces several novel methods not assessed in the literature, deemed "Experimental methods", and tested on the validation set to determine whether they improve forecasting skill.

Per-Organization Mode Differencing All statistical models (ridge regression, RF,

ET, and XGBoost) are trained not on the raw ratings y_i but on per-organization residuals $\tilde{y}_i = y_i - m_o$, where m_o is the mode of the training ratings for reporting organization o. The per-organization mode m_o is computed from the training set only and is used as a fixed offset for all activities from that organization. Predictions from each model are therefore organization-relative residuals; the final predicted rating is recovered by adding back m_o:

$$\hat{y}_i = \hat{\tilde{y}}_i + m_o.$$

This decomposition removes the dominant between-organization rating shift before model training, allowing models to focus on within-organization variation. The per-organization mode also serves as the mode baseline described in Section 2.8.1.

Nearest Neighbor (Vector Similarity) A similarity measure was first constructed from features including the activity's countries, GDP per capita, scope, and implementing and funding organization ID. This measure, however, significantly underperformed semantic similarity derived from the *gemini-2.5-flash*-generated activity summary, so semantic similarity was used instead. Similarity was weighted proportional to its embedding semantic similarity score, and a cutoff was tested for averaging 1, 3, 7, 10, 15, and 20 nearest neighbors using the Gemini embeddings model *gemini-embedding-001*, embedded on the activity title and LLM-generated summary (in this case not on the "targets" text). 15 nearest neighbors was found to be the highest-performing number. The weighted average of the nearest neighbor ratings was however not found to be predictive of the overall activity score and was not incorporated as a feature or prediction method.

Although the nearest-neighbor method was used to collect examples for the LLM prompt, simply taking the weighted mean of 15 similar ratings performed worse than the "most common rating" method and was not adopted as an experimental method to directly forecast ratings.

XGBoost The XGBoost (Extreme Gradient Boosting) method is a statistical algorithm which constructs an ensemble of decision trees sequentially, where each subsequent tree attempts to correct the errors made by the previous trees. Unlike the RF which trains trees independently and averages their predictions, XGBoost builds trees iteratively, with each new tree focusing on the residual errors of the ensemble.

The algorithm incorporates both L1 and L2 regularization terms to penalize model complexity and prevent overfitting on the training data. The decision trees split based on value ranges of the features, using a splitting criterion that accounts for the gradient and Hessian of the loss function.

LLM Adjustment Ridge Regression The goal of the LLM Adjustment was to incorporate usable information from the direct LLM forecasts, since simply averaging the LLM forecasts with the statistical model in the validation set decreased performance. To do so, a small ridge regression model was layered on top of the forecasts, calibrated using

LLM predictions on the training set.

Let $\hat{y}_i^{\mathrm{RF}}$ be the RF+ET prediction for activity i, and let $\hat{y}_i^{\mathrm{LF}}$ denote the LLM Forecast. Note that the RF prediction is averaged with the ET model but is still referred to as "RF" in the equations. The RF+ET residual for activity i is defined as:

$$r_i := y_i - \hat{y}_i^{\mathrm{RF}}.$$

A ridge regression model was then fitted to predict residuals from a small feature vector consisting of the RF+ET prediction and (optionally) the LLM Forecast:

$$\hat{r}_i := \beta_0 + \beta_1 \hat{y}_i^{\mathrm{RF}} + \beta_2 \hat{y}_i^{\mathrm{LF}},$$

with an ℓ_2 penalty on (β_1, β_2) controlled by $\alpha = 5$, the ridge strength. The corrected prediction is:

$$\hat{y}_i^{\mathrm{corr}} := \mathrm{clip}_{[1,6]}\left(\hat{y}_i^{\mathrm{RF}} + \lambda \hat{r}_i\right),$$

where λ is a scaling factor (set to 1.0 in the experiments) and clipping enforces the valid rating range between 1 and 6.

The ridge model takes both the RF+ET prediction and the LLM Forecast as inputs for activities where $\hat{y}_i^{\mathrm{LF}}$ is available (the training activities for which LLM forecasts were generated, spanning the full validation set). This allows the correction to learn a mapping from $(\hat{y}_i^{\mathrm{RF}}, \hat{y}_i^{\mathrm{LF}})$ to the residual r_i. The model effectively learns the degree to which each input should shift the prediction from the mean, together with a fixed learned offset. The only learned parameters are β_0, β_1 and β_2. The ridge regression LLM adjustment improved validation set performance and was adopted for testing against a simple average on the test set.

2.9 LLM Forecasting of Overall Activity Success

Multiple studies have measured zero-shot LLM forecasting capability (zero-shot means without any complex prompt architectures or examples specific to the forecasting task), and found that better general ability base models tend to perform better on forecasting tasks (Halawi et al., 2024) (Karger et al., 2024): In one study with dozens of base models and a dynamically updating benchmark on prediction market forecasting questions, an inverse linear relationship was found between the human preference of a model's answer (in terms of an ELO score) and the Brier score (Karger et al., 2024). A similarly inverse log-linear relationship was found between the compute used to train the model and the Brier score, both demonstrating increased forecasting skill with model size. Several LLM forecasting methods were investigated to improve overall rating forecasts and narrative forecasting of outcomes.

2.9.1 LLM Forecasting Baseline Methods

***deepseek-R1* Provided Only the Activity Summary and Statistics** Several LLM methods are introduced, closely informed by prior state-of-the-art methods in judgemental forecasting literature (Halawi et al., 2024) (Lee et al., 2025). To justify these methods, comparison is made to a simple baseline where all that is provided to the model is the activity ID, the activity title, an activity summary from *gemini-2.5-flash* of the 10 most important pages from the activity documents for forecasting the outcome, and the statistical prevalence of each outcome rating. The model is then asked directly to forecast the evaluation rating.

***gemini-2.5-Flash* Summary of Risks** A separate baseline used only for assessing the forecast grades of models, is submitting a summary of the risks identified from activity documents to the *gemini-2.5-flash-lite* grading LLM, and comparing these to proper LLM forecasts. This also provides a sanity check that the LLM is not simply summarizing what the authors of activity of appraisal documents already highlight as risks. However, the lack of assertive claims about what will happen in a positive sense hinders this baseline metric relative to graded forecasts.

2.9.2 LLM Forecasting Experimental Methods

The LLM forecasting method was decided upon by iteratively inspecting both the quality of the response, and the overall accuracy of the forecasts made by the LLM. *gemini-2.5-pro* generated a series of "mock forecasts" to provide examples of high-quality forecasts for similar activities to the forecasting model.

To generate each mock forecast, a retrieval-augmented input was constructed consisting of up to 10 pre-activity pages and up to 10 post-activity evaluation pages per activity. Pre-activity pages were selected from high-scoring pages across 19 of the 23 pre-activity sub-categories (objectives, implementation plans, risks, financing details, contextual challenges, and stakeholder/implementer information; excluding condensed summaries, quantitative targets, budget and legal pages, and monitoring detail pages). A minimum relevance score of 9/10 was required for pre-activity pages, leaving only highly relevant pages. For pages that did match, nearby adjacent pages were also included when insufficient high-scoring pages were available to prevent relevant information and informative tables from being cut off.

Pages were selected from post-activity documents which emphasized deviations from plans (including deviations, delays/early completion, and over/under-spending), using a lower relevance threshold (minimum score of 3) and likewise included surrounding pages to reach the target count when needed. These retrieved excerpts were then merged with activity metadata (title, scope, planned start/end dates, planned financing totals when present)

and brief model-generated pre-activity summaries (activity description and risk summary). Gemini was then prompted to write a forecast from the ex-ante perspective. Importantly, the prompt required the model to end by outputting the *known* final evaluation rating for that activity (derived from the merged ratings file and converted into scale-specific text via `get_ratings_text`). The prompt also instructed the model to ground the narrative in the retrieved evaluation pages and to return "NO RESPONSE" if the evaluation excerpts did not contain sufficient justification for the assigned rating.

The most semantically relevant activities which ended approximately at or before the start of the activity being forecast were then retrieved (see Section 2.8.2). In addition, the activity "risks" grade from *gpt-3.5-turbo* was inserted before each mock forecast, to provide context for the example. Each mock forecast was structured similarly to the highest-performing scratchpad method from (Halawi et al., 2024). The activity title, start date, location, and a *gemini-2.5-flash*-generated summary were also injected into the prompt.

Finally, the distribution of rating outcomes was inserted into the prompt, to prevent collapse towards only a few ratings.

Prompt Template and Multi-stage Prompting The full prompt is composed of the following blocks, assembled in the order listed:

1. **Activity metadata** — title, planned start/end dates, scope, planned disbursement (USD), location(s), GDP per capita, reporting organisation.
2. **Pre-activity description & targets** — a *gemini-2.5-flash*-generated activity description, targets summary, contextual challenge description, complexity details, integratedness description, financing summary, and implementer-performance context.
3. **Risk summary** — a *gemini-2.5-flash*-generated structured summary of key project risks.
4. **KNN few-shot block** — summarised mock forecasts of the three most semantically similar training activities, one from each broad rating tier (`{knn_summary_text}`) (see Section 2.8.2).
5. **Retrieved pre-activity excerpts (RAG)** — additional activity-specific information retrieved by the model from the available documents and synthesised into a short passage (`{rag_synthesis_additional_info}`).
6. **Multi-stage reasoning (S1/S2)** — outputs of two prior LLM calls: reasons the activity may have been rated "Moderately Satisfactory" or worse (S1) and reasons it may have been rated "Satisfactory" or "Highly Satisfactory" (S2). This left approximately 50% of responses above and below the threshold for consideration.
7. **Rating distribution prior** — a short description of the empirical distribution of rating outcomes in the training data, prepended to the scratchpad method

instruction block (not appended as the final block).

The full prompt template for the LLM Forecast is shown in Figure 2.9.2. LLMs in all prompts were instructed to "think like a superforecaster" (Tetlock and Gardner, 2015). Superforecasters typically outperform domain experts by maintaining broad judgement and cultivating calibrated probability estimates, rather than relying on a single strategy such as econometric analysis. This replicates prior work finding this instruction to be the highest-performing prompt strategy in the geopolitical domain (Halawi et al., 2024).

In both methods, a k-nearest-neighbors (KNN) few-shot block of semantically similar training activities is used (see Section 2.8.2 for how semantic similarity was determined). The language model was prompted to extrapolate lessons about rating scales from the most similar example in each rating tier: the most similar "Highly Unsatisfactory", "Unsatisfactory", or "Moderately Unsatisfactory" example; the most similar "Moderately Satisfactory" example; and the most similar "Satisfactory" or "Highly Satisfactory" example. A selection of $k = 3$ summarized mock forecasts was found to perform better than $k = 1$, 5, or 7.

Each example activity in the few-shot block included (i) key metadata (title and, where available, location and a brief summary), (ii) a short "risks" summary, (iii) the retrospective mock forecast analysis, and (iv) the final evaluation outcome label.

Finally, a short description of the empirical distribution of rating outcomes in the training data was appended. This was found to reduce mode-collapse toward a narrow subset of ratings.

Box 2.1: Single-method multi-stage forecasting prompt

Stages s1 and s2 are run as separate calls, and their outputs are inserted into the final (s3) prompt via `insert_stage_s1_answer_here` and `insert_stage_s2_answer_here`. Below are the system and user prompts.

```
SYSTEM:
You are an experienced international aid decision maker with a quantitative
mindset.  Respond with a comprehensive, thorough forecast of what the overall
evaluation rating of the activity will be, from the options of {options_text}.

USER:
Forecast what the outcome will be for this activity.

### Lessons from similar activities ###
{knn_summary_text}
### End lessons ###

### Additional specific information about the activity that you summarized ###
{rag_synthesis_additional_info}
### End of additional information you summarized ###

ACTIVITY ID: {activity_id}
ACTIVITY TITLE: {activity_title}
```

```
ORIGINAL PLANNED START DATE: {planned_start}
ORIGINAL PLANNED END DATE: {planned_end}
ACTIVITY SCOPE: {activity_scope}
PLANNED TOTAL DISBURSEMENT (USD): {planned_total_disbursement_usd}
ACTIVITY LOCATION(S): {locations}
LOCATION GDP PER CAPITA, USD: {gdp_percap}
PARTICIPATING ORGANIZATIONS: {reporting_orgs}
IMPLEMENTING ORGANIZATION CATEGORY: {either "Government" or "NGO", otherwise
line not inserted}

ACTIVITY DESCRIPTION: {ChatGPT_description}
ACTIVITY TARGETS: {targets_summary}
ACTIVITY CONTEXT: {activity_context}
ACTIVITY COMPLEXITY: {complexity_details}
ACTIVITY INTEGRATEDNESS: {how_integrated_description}
FINANCING DETAILS: {finance_summary}
IMPLEMENTER PERFORMANCE CONTEXT: {implementer_performance_text}
ACTIVITY RISKS:
{risks_summary}
ACTIVITY POSSIBILITIES: {possibilities_summary}

{training-set rating distribution text}
Here are a few reasons that you said the answer might be "Moderately
Satisfactory" or worse:
{insert_stage_s1_answer_here}
Here are a few reasons that you said the answer might be "Satisfactory" or
better:
{insert_stage_s2_answer_here}

YOUR TASK:
Aggregate your considerations above.  Think like a superforecaster (e.g. Nate
Silver).  On the very last line of your response, write 'FORECAST: ' followed by
exactly one option from this rating scale with no extra words:
{options_text}

Respond only in English.
```

Inference-time Enhancements (Ensembling) Ensembling averages many individual forecasts of the same model, prompted in slightly different ways. Ensembling was found to be a weak method when running on 100 sample activities within the training set, because the same model+prompt pair had highly correlated forecasts. Creating a new prompt variant for each ensemble run was also not attempted due to the complexity of such an approach, its high API cost, and weak expected performance improvements.

Model Adaptation (Fine-tuning / DPO) In past work in a similar domain, fine-tuning significantly improved forecasting performance (Wen et al., 2025). An attempt was made to fine-tune *gemini-2.5-flash* using Vertex AI. To do so, Direct Preference Optimization (DPO) was used, which requires one example of a good prompt, and one example of a bad prompt. Out of a random sample of 100 activity IDs in the training data, the final forecast stage was forecast 5 separate times using *deepseek-R1*. 50 pairs were used where the model forecast one rating increment closer than another rating to the true label.

Because there were often multiple equally good or equally bad options on the limited 6-point scale, the embeddings of the forecasts were taken and cosine similarity was computed

to the embeddings of the mock forecast and to embeddings of the post-activity document; these similarity scores were then averaged. The most similar among equally good ratings were chosen as the good example for the DPO training pair, and the least similar among equally bad ratings as the bad example.

Once the 50 pairs were identified, fine-tuning was conducted using default settings over 20 epochs from Vertex AI (Learning rate multiplier of 1, Adapter size of 4, Beta of 0.1).

2.10 Forecasting Cost-Effectiveness with RF+ET

The z-score was the metric chosen for training and testing the statistical model on cost-effectiveness forecasting. The z-score is the average of all quantitative outcomes divided by their share of the total expenditure of the activity. The mean in the training set is then subtracted and the result is divided by the standard deviation, across all categories. As a result, the cost-effectiveness for each metric is always a similar judgement for the model. The model is trained on these z-scores, where all z-score predictions are averaged together to determine overall accuracy. A single RF model was used for z-score predictions and training.

2.10.1 Cost-effectiveness Baseline Methods

Because the model struggled to achieve above-chance performance, the only baseline was a 50% pairwise ranking skill, and an R^2 of 0, equivalent to random assignment of cost-effectiveness among activities, and always predicting the mean value of the test set respectively.

2.10.2 Cost-effectiveness Experimental Methods

The RF+ET ensemble was used for cost-effectiveness forecasting, using the same 59 features as the overall rating model, plus an additional 7 dummy variables to indicate which of the outcomes are being forecast (one for each outcome category, with one for each unit of CO_2). The models were trained with 3× sample weight on the most recent 20% of training observations, as this improved performance on the validation set. A ridge regression on the residuals as a function of start date with $\alpha = 50$, as used for outcome tags and overall rating prediction, was also applied due to its performance improvement on the validation set. Otherwise, cost-effectiveness model training was identical to that of overall ratings. $\alpha = 50$ was selected to encourage a conservative time dependent ratings change, rather than a strong start date dependence on ratings.

2.11 Outcome Tag Forecasting

2.11.1 Outcome Tag Baseline Methods

Constant Base-Rate Prediction for Outcome Tags For binary outcome tags, the no-skill baseline predicts the training-set positive rate $\bar{p}_{\text{train}}$ for every activity, regardless of its features. This constant forecast is the natural reference for the Brier skill score: a model that achieves BSS > 0 beats this baseline, while BSS $= 0$ indicates the model adds no probabilistic skill over simply forecasting the base rate. The same constant baseline also sets the accuracy of a no-feature classifier: a classifier that ignores all inputs and always predicts the majority class achieves accuracy $\max(\bar{p}_{\text{train}}, 1 - \bar{p}_{\text{train}})$. The accuracy ratio reported for each tag (Table 4) is therefore the model accuracy divided by this constant-baseline accuracy, with values above 1 indicating improvement.

2.11.2 Outcome Tag Experimental Methods

For each outcome tag, an RF+ET ensemble classifier was trained using the same initial feature set and time-ordered split as the overall rating model (see Table 1 and Figure 2).

A tag's model was deemed ineligible for test-set prediction unless the Brier skill score was strictly positive and the within-group pairwise ranking probability exceeded 50%. Otherwise, the model was reverted to the constant baseline, due to low confidence that the RF+ET ensemble would generalise beyond the validation set.

The chosen 14 forecast tags were:

- **Funds Cancelled or Unutilized (unsigned):** Project funds were cancelled, not disbursed, or significantly underutilized.
- **Funds Reallocated (unsigned):** Project funds were reallocated across components or activities.
- **High Disbursement (unsigned):** Project achieved high or full disbursement of funds.
- **Improved Financial Performance (unsigned):** Improved revenues, profitability, financial management, or financial capacity of beneficiary institutions were significantly achieved.
- **Project Restructured (unsigned):** Project was formally restructured, including changes to scope, components, or objectives.
- **Closing Date Extended (unsigned):** Project closing date was extended beyond the original plan.

- **Targets Revised (unsigned):** Project targets, indicators, or results framework were revised during implementation.
- **Policy and Regulatory Reforms (signed):** Policy, regulatory, or legal frameworks were significantly and successfully developed, strengthened, or operationalized.
- **Targets Met or Exceeded (signed):** Project targets were met as planned, or exceeded.
- **Over Budget (signed):** Project had to exceed the original budget.
- **Capacity Building Delivered (signed):** Capacity building, training, or institutional strengthening activities were significantly delivered.
- **High Beneficiary Satisfaction or Reach (signed):** High beneficiary satisfaction, broad beneficiary reach, or high uptake was achieved.
- **Gender-Equitable Outcomes (signed):** Outcomes affecting women and girls were equitable or positive.
- **Private Sector Engagement (signed):** Success in attracting private sector investment or improving the business environment was achieved at or above targets.

Tags excluded from test-set prediction due to poor validation performance were: Monitoring and Evaluation Challenges, Energy Sector Improvements, Improved Livelihoods, Improved Service Delivery, Infrastructure Completed, Activities Not Completed, Design or Appraisal Shortcomings, External Factors Affected Outcomes, and Implementation Delays. Many of these had very few negative examples or a highly imbalanced true/false split, which led to weak model performance.

Outcome Tag Feature Selection An ablation experiment for all tags was conducted on the validation set. The RF+ET model uses 59 features, which may not all be appropriate for every outcome tag; a series of experiments was conducted to determine the best feature subset per tag.

Features were grouped into LLM grades, governance indicators, regions, budget, organisational structure, UMAP features, sector clusters, economic context, and missingness flags, and assessed on a per-tag basis. Using all non-ablated features for every tag reduced model performance on the validation set. Some tags were unrelated to many features, and the imbalanced true/false split combined with sparse tag data meant that effective training sets were much smaller for some tags, increasing the risk of overfitting.

A feature-selection sweep tested four strategies: all 45 features (no reduction) and subsets of the top-5, top-10, and top-30 features ranked by RF feature importance on the training set. To select among these candidates without capitalising on a chance-best result, a

pairwise tournament across Brier skill score, accuracy ratio, and within-group pairwise ranking skill was used.

Small losses on one metric (below 1.5 percentage points) were forgiven if a compensating gain on another metric was at least 20% of the loss. The candidate with more net wins was selected; ties were broken by Brier skill score.

Because this selection was made on the same validation set used to evaluate final performance, there was a risk that the feature reduction reflected overfitting to that split rather than a genuine regularisation benefit. To check this, the selected feature set for each tag was re-evaluated using three-fold temporal cross-validation on the training set alone, where the validation set played no role.

Because it was unclear whether aggressive per-tag feature reduction or a more permissive feature set would generalise better, both the limited per-tag feature set and the full 45-feature set were used for test-set forecasting.

A group-level ablation additionally revealed that governance indicators, engineered features (mean World Governance Indicator, CPIA score, expenditure times complexity) and missingness-flag features were introducing more noise than signal. These 14 features (2 governance, 11 missingness, 1 feature engineering) were dropped from all tags, leaving 45 features available for per-tag selection. Removal improved scores on the k-fold temporal cross-validation on the training set. For each tag assigned a reduced feature subset, k-fold performance was at least as good as the full-feature alternative, indicating that the reduction reflects a genuine generalisation benefit.

RF+ExtraTrees Ensemble RF and ET both used classifier models rather than regressors as in the ratings training, as the method is intended for binary probabilistic prediction rather than a regression prediction. OLS linear regression, XGBoost, and LightGBM were tested against the RF+ET model, but early experiments showed they performed significantly worse on all evaluation metrics so use of these models was not continued.

Three options were evaluated on the validation set: The `class_weight` parameter was set to always `none`, always `balanced`, and a tag-dependent scheme. The tag-dependent scheme selected a threshold where above 65% of training examples were positive, `class_weight=None` was used allowing the model to predict near the natural base rate, and below 65% `class_weight="balanced"` was used to prevent the ensemble from collapsing to the majority class. The tag-dependent scheme was selected for the test set evaluation, as it performed better than the other two options.

Optimal parameters for the RF+ET classifier ensemble were identified by exploring 20 LLM-generated parameter sets, with the second set of 10 varied near the maximal score of the first set. These included variations in tree depth, leaf size, feature sampling width,

row subsampling, and pruning strength, as well as between-RF and ET variations.

The base configuration (`min_samples_leaf`=5, `max_features`="`sqrt`", `n_estimators`=500, `max_depth`=`None`, `min_samples_split`=2, `ccp_alpha`=0), with bootstrap `True` for the RF and `False` for the ET, was found to perform best across most tags on the validation set. Ten tags use per-tag hyperparameter overrides identified during the parameter sweep: five tags (high disbursement, policy and regulatory reforms, capacity building delivered, private sector engagement, and gender-equitable outcomes) use `min_samples_leaf`=20; one tag (over budget) uses `min_samples_leaf`=40; one tag (funds reallocated) uses `min_samples_leaf`=10; and three tags (high beneficiary satisfaction or reach, targets met or exceeded, and monitoring and evaluation challenges) use `max_depth`=10. Although quite similar to the default `RandomForestClassifier` and `ExtraTreesClassifier` configuration, this specification constrains model capacity to reduce overfitting on the small training sets involved.

The RF+ET ensemble was otherwise trained identically to how ratings were trained.

Tag Group Averaging Analysis of outcome tags in the training data revealed that certain groups of tags are strongly correlated. For example, tags relating to targets being met correlate with good financial performance. Three factor groups were identified in the data; two are used for direct blending and a third contributes indirectly:

- Success Group: average of six outcome tags (targets met, beneficiary satisfaction, private sector engagement, capacity building, policy reforms, and financial performance).
- Financing Group: disbursement tag, funds-cancelled tag (inverted).
- Rescoping Group: Project Restructured, Targets Revised, Closing Date Extended, Funds Reallocated. This group's RF+ET regressor is trained and its predicted score is used as an OLS predictor input when reconstructing per-tag probabilities for blended tags, but rescoping-group tags are not directly blended (the group influences other tags via OLS rather than being a direct mixture target).

For each group, an RF+ET regressor was trained on the 45 features to predict the group score. The per-tag probabilities were then reconstructed from predicted group scores via an OLS linear regression fitted on training data.

It was observed that the group average probability was typically more informative for low-data tags. The minority class count is defined as the minimum between the number of true or false tags in the training data. 50 different blend selection options were swept across, varying the minority-class training point threshold and the mix ratio of the group and tag average above that threshold.

The best performing blend on the validation set was a linear increase from 100 to 400 minority-class training points:

$$\hat{p}_i^{\text{final}} = w \cdot \hat{p}_i^{\text{model}} + (1 - w) \cdot \hat{p}_i^{\text{group}},$$

where $w = \text{clip}\left(\frac{m-100}{400-100},\ 0,\ 1\right)$ and m is the minority class count. Below 100 training points the group average is used exclusively; above 400 the per-tag model prediction is used exclusively.

LLM Outcome Tag Forecast An LLM-based correction step was tested for tag forecasting, analogous to the approach described for the overall rating model. For each of the 14 tags whose statistical model beat the constant baseline, *deepseek-R1* with thinking enabled was prompted with the activity title, an LLM-generated summary of the pre-activity documents, the tag label and definition, and the training-set base rate. The model was instructed to reason like a superforecaster: enumerating reasons the outcome was likely true, then likely false, then anchoring on the base rate, before producing a final probability. Forecasts were produced for 150 random training activities and all validation activities. A trial of blending LLM probabilities with statistical model probabilities was attempted. A sweep of 20 blend weights in increments of 5% from 0% to 100% LLM was conducted, measuring mean within-group ranking skill, accuracy, and Brier skill score across the 14 non-baseline tags on the validation set.

Mean accuracy declined monotonically with LLM weight; within-group ranking skill and Brier skill showed only marginal gains at 5–10% LLM before deteriorating; at 70% LLM the mean Brier skill score turned negative. The LLM correction was therefore discarded and the statistical model predictions are used as the final tag forecasts.

2.12 Scoring Metrics

Each metric used to evaluate forecasts is described below. Throughout, N denotes the number of observations in the evaluation set, y_i the true label for observation i, $\hat{y}_i$ the corresponding prediction, $\hat{p}_i \in [0, 1]$ the corresponding predicted probability, and $\mathbf{1}[\cdot]$ the indicator function (equal to 1 when its argument is true and 0 otherwise).

Accuracy The percent of the time the correct rating is forecast:

$$\text{Acc} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\text{round}(\hat{y}_i) = \text{round}(y_i)].$$

Non-integers are rounded to integers for overall ratings. While accuracy has the advantage of intuitive simplicity, it is not strictly proper (Gneiting and Raftery, 2007), meaning that maximizing the model accuracy may not maximize a model's reported true belief about the probability of an outcome. This can be problematic if used to select between models.

RMSE (Root Mean Square Error) Take the square of the difference between every prediction and the true value, take the mean of all such squared values, then take the square root:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2}.$$

This is a measure of "average" distance. Lower is better. As ratings range from 1 to 6, the difference is on a scale from 0 to 5, where the worst possible value is 5, best possible value is zero. This method heavily penalizes predictions that are significantly incorrect. RMSE is only used in this work for overall ratings.

MAE (Mean Absolute Error) A measure of "average" distance, calculated by taking the mean of the absolute value of the residuals:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}_i - y_i|.$$

Lower is better. As ratings range from 1 to 6, a scale from 0 to 5 is best, therefore the worst possible value is 5, and the best possible value is zero. In the case of overall success ratings, this metric tends to reward integer rounded rating scores as they are often exactly correct, rather than the output of the statistical models which are not exact integers. This method does not heavily penalize predictions that are significantly incorrect. MAE is only used in this work for overall success ratings.

Coefficient of Determination (R^2) R^2 measures the proportion of the variation in the dependent variable that is predictable from the independent variable:

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2},$$

where $\bar{y}$ is the mean of the true labels. Higher is better. Theoretically, R^2 equals zero when always predicting the mean of the evaluated true labels. Using the training set mean results in a negative score on the test set. If more than 1 regressors are included, R^2 is the square of the coefficient of multiple correlation and can be negative. This metric often correlates strongly with RMSE. R^2 is only used in this work for overall ratings and cost-effectiveness.

Adjusted R^2 Adjusted R^2 is a version of R^2 that accounts for the number of regressors in the model, and is well-defined for OLS linear regression:

$$\bar{R}^2 = 1 - (1 - R^2)\frac{N-1}{N-k-1},$$

where k is the number of regressors. Unlike plain R^2, adjusted R^2 penalizes adding predictors that do not meaningfully improve fit. It can decrease when irrelevant regressors

are included. A higher adjusted R^2 is better. While it penalizes extra parameters that may lead to overfitting, adjusted R^2 within a training set does not reflect model skill as accurately as out-of-time R^2. Adjusted R^2 is reported in this work on the OLS linear regression for comparison with prior literature. Adjusted R^2 is only used in this work for overall ratings.

Pearson Correlation Pearson's correlation coefficient measures the strength and direction of linear association between two variables X and Y:

$$r_{X,Y} = \frac{\mathrm{cov}(X,Y)}{\sigma_X \, \sigma_Y},$$

where $\mathrm{cov}(X,Y) = \frac{1}{N}\sum_{i=1}^{N}(X_i - \bar{X})(Y_i - \bar{Y})$ is the covariance and σ denotes the standard deviation. The ratio is unitless and lies in $[-1,+1]$: $+1$ indicates a perfect positive linear relationship, -1 perfect negative, and 0 no linear association. Its square r^2 is the fraction of variance in one variable explained by a linear function of the other, and coincides with R^2 in linear regression with one predictor variable. It is used as an evaluative metric to assess how forecasts correlate with each other and other variables.

Within-Group Pairwise Ranking Within-group pairwise ranking is evaluated as the mean across groups of the proportion of within-group pairs of predictions that were correctly ordered:

$$\mathrm{WGPR} = \frac{\sum_g n_g \, \mathrm{PR}_g}{\sum_g n_g}, \qquad \mathrm{PR}_g = \frac{\sum_{(i,j)\in g,\, y_i \neq y_j} \mathbf{1}[(\hat{y}_i - \hat{y}_j)(y_i - y_j) > 0]}{\sum_{(i,j)\in g,\, y_i \neq y_j} \mathbf{1}[\hat{y}_i \neq \hat{y}_j]},$$

where n_g is the size of group g. Only pairs of activities sharing the same reporting organisation and start year are compared; between-group pairs are excluded entirely. For each such pair (i,j) where the true outcomes differ ($y_i \neq y_j$), the prediction is counted as correct if $\hat{y}_i > \hat{y}_j$ whenever $y_i > y_j$, and incorrect otherwise; pairs where $\hat{y}_i = \hat{y}_j$ are excluded from both numerator and denominator. Each group's score is computed independently and the final metric is the weighted mean across groups proportional to the number of activities in each group, so a large group does not dominate a small one. Restricting comparison to activities from the same organisation and group year removes systematic between-group variation in base rates that a model cannot be expected to predict, making the metric a cleaner measure of within-context discriminative skill. This method is insensitive to global shifts in ratings or probabilities, which may occur due to events like the COVID-19 pandemic or differences in organisational reporting culture, and insensitive to calibration of the spread of predicted probabilities. A score of 0.5 corresponds to random chance; 1.0 is perfect ordering. If the tied variables were the more difficult to rank, then WGPR may over-represent the skill of the model as the ties are excluded.

Within-Group Spearman Ranking Within-group Spearman rank correlation applies

Pearson's correlation to rank vectors within groups sharing the same reporting organisation and start year:

$$\rho_g = r_{\mathrm{rank}(\hat{y}_g),\,\mathrm{rank}(y_g)}, \qquad \mathrm{WGSR} = \frac{\sum_g n_g \rho_g}{\sum_g n_g},$$

where rank$(\cdot)$ denotes midranks within group g: the average of the positions a value would occupy if ties were broken arbitrarily, so that tied values receive the average of the ranks they would have occupied. The reported score is the weighted average of per-group ρ values, with weights proportional to group size. Like WGPR, this metric controls for between-group differences in base rates, but unlike WGPR it captures the degree of monotonic ordering rather than merely whether the direction of each pair is correct.

Brier Score The Brier score, used here in its binary form, is the mean squared error between the predicted probability and the binary outcome $y_i \in \{0, 1\}$:

$$\mathrm{BS} = \frac{1}{N} \sum_{i=1}^{N} (\hat{p}_i - y_i)^2.$$

Lower values are better, with 0 indicating perfect probabilistic forecasts and 1 being the worst possible. The Brier score is a strictly proper scoring rule (Gneiting and Raftery, 2007), meaning that a forecaster maximises their expected score only by reporting their true beliefs. Only the binary Brier score is used in this work for outcome tags forecasts, as they are binary and probabilistic.

Brier Skill Score The Brier skill score (BSS) measures improvement over a reference forecast:

$$\mathrm{BSS} = 1 - \frac{\mathrm{BS}}{\mathrm{BS}_{\mathrm{ref}}},$$

where $\mathrm{BS}_{\mathrm{ref}}$ is the Brier score of the constant (no-skill) baseline that always predicts the training-set base rate. A BSS of 0 indicates no improvement over this baseline; positive values indicate skill. The Brier skill score is only used for outcome tags forecasts, as they are binary and probabilistic.

2.12.1 Choosing Metrics and Methods for the Test Set

Chosen Quantitative Forecast Evaluation Metrics Before evaluating on the test set, one must choose the metrics to evaluate in advance. This prevents post-hoc metric selection (choosing the most favorable metric after examining the results). After consideration of several metrics, the within-group pairwise ranking and R^2 were determined to be the most appropriate for assessing model skill in the domain of ratings and activity quantitative outcomes, while other metrics still provide useful insight. R^2 is sensitive to bias and outliers, which are important for assessing absolute predictive accuracy. However, a common use-case in aid funding decision making is comparing a pair of activities, or even

a suite of many activities. In such a use-case, the within-group pairwise ranking is more representative of what is needed. Furthermore, a weakness in R^2 is that global shifts in the ratings and outcomes due to external factors can be unpredictable, and a model may by-chance capture these shifts without any real improvement in forecasting skill. Both are clearly interpretable: always forecasting the mean value of the test set is equivalent to an R^2 of 0, also meaning that 0% of variance explained for R^2, while 50% is equivalent to the within-group pairwise ranking forecast of random chance. R^2 and within-group pairwise ranking were also applied to LLM forecasts of overall outcome ratings. For outcome tags, the primary metric was within-group pairwise ranking for comparability to overall rating prediction.

Chosen Qualitative Forecast Evaluation Metric In addition to extracting the LLM forecast rating, *gemini-2.5-flash-lite* was also used to grade narrative textual format forecasts. For all ensembles of LLM forecasts, a grade was given for how similar the forecast was to the activity post-activity evaluation summary using *gemini-2.5-flash-lite*. Grading was performed based on two key criteria: accurate identification of likely drivers of activity success or failure, and identification of likely outcomes being forecast, which together inform the single combined letter grade.

In accordance with the typical US grading scheme definitions, an "F" grade is defined as 55 or lower and an A+ grade as approximately 97, with other grades defined in approximately even intervals, alternating between 3-point gaps within sub-grades and 4-point gaps between main letter grades. For example, D-=62, D=65 (3-point), and D+=68 to C-=72 (4-point). *gemini-2.5-flash-lite* was provided the following rubric:

Grading Scale

- A+/A/A-: Excellent forecast, highly accurate, attention on key drivers, multiple major events forecast accurately.
- B+/B/B-: Good forecast, mostly accurate. A mix of correct and incorrect, but at least one major outcome was forecast. At least one key driver identified.
- C+/C/C-: Adequate forecast, partially accurate or reasonable. Focus was adequate. Major outcomes incorrect, but some smaller aspects were correct.
- D+/D/D-: Poor forecast, although perhaps one or two small correct predictions. Mostly inaccurate. Wrong focus on drivers.
- F: Failed forecast, completely wrong or unsupported.

Methods Chosen for the Test Set To ensure that methods are not chosen specifically to look good on the test set, the selected methods were determined in advance as follows:

- **Statistical Models for Overall Ratings:** As baseline methods I chose to assess the performance of always predicting the most common rating within the reporting org (*Mode of reporting-org score baseline*), a simple ridge regression model with

only the overall risk LLM grade and the reporting org (*Ridge Baseline (risks + org only)*) as a means of approximating what human evaluators may have assigned the likely overall rating, and a simple linear model with all features (*Plain OLS GLM*) to assess the importance of nonlinearity for the statistical model performance on the test set. My primary method was (*RF+ET all features + year corr*) which was the average of the RF and ET model after adjusting year correlation, and I assessed the following alterations to see if they improved performance: the RF+ET ensemble averaged with the LLM forecast (*RF+ET + LLM Forecast (simple avg)*), the more sophisticated calibrated ridge regression to combine the statistical model with the LLM forecast *RF+ET + LLM Forecast (ridge)*, removing the linear year correlation adjustment (*RF+ET all features (no year corr)*), removing LLM input features from the model (*RF+ET, no LLM features*), resetting the RF+ET tuned parameters to their default values (*RF+ET (default params)*), and replacing the RF+ET with the XGBoost nonlinear model (*XGBoost all features*). Final results on the test set are reported in the results Section 3.2.

- **LLM Forecasts** The best performing model was the *deepseek-R1* baseline minimal prompt, which was evaluated on the test set.
- **Cost-Effectiveness** Due to poor validation performance, only the standard RF+ET with year correlation was chosen to be assessed.
- **Outcome Tags** Only two methods were assessed on the test set for outcome tags: including the per-tag features and parameters (all subsets of the 45 "non-noisy" features, where each per-tag parameter set was a less complex model), and separately using always the same 45 "non-noisy" features which did well on the validation set.

2.13 Summary

A dataset of 1181 completed and evaluated international aid activities from the IATI dataset with matched sets of pre-activity descriptions and post-activity evaluations was extracted. To ensure temporal leakage does not occur, thousands of PDFs were downloaded, ranked from most to least relevant for forecasting future outcomes or evaluating the end result of the activities, had their pages graded for relevance to the task, and had quantitative and qualitative descriptions and results extracted from the PDFs. Three time-ordered splits were used: train, validation, and test. Statistical models and fine-tuning experiments were done on the training set, with several methods assessed using the validation set. This informed model selection and parameter tuning, so that the generalization ability of these methods were finally assessed against the test set. Specifically, four evaluative approaches for activity success forecasting were attempted:

- A statistical model to forecast overall activity success ratings using LLM-generated features to be assessed on R^2 and within-group pairwise ranking skill.
- An LLM prompting scheme used to forecast overall activity success ratings, assessed on R^2, within-group ranking skill, and qualitative similarity to the evaluation summaries, optionally incorporating statistical overall rating forecasts as a target.
- A statistical model to forecast activity cost-effectiveness using LLM-generated features to be assessed on R^2 and within-group pairwise ranking skill.
- A statistical model to forecast 14 binary outcomes using LLM-generated features to be assessed on within-group pairwise ranking skill.

See Figure 5 for a representation of the entire process.

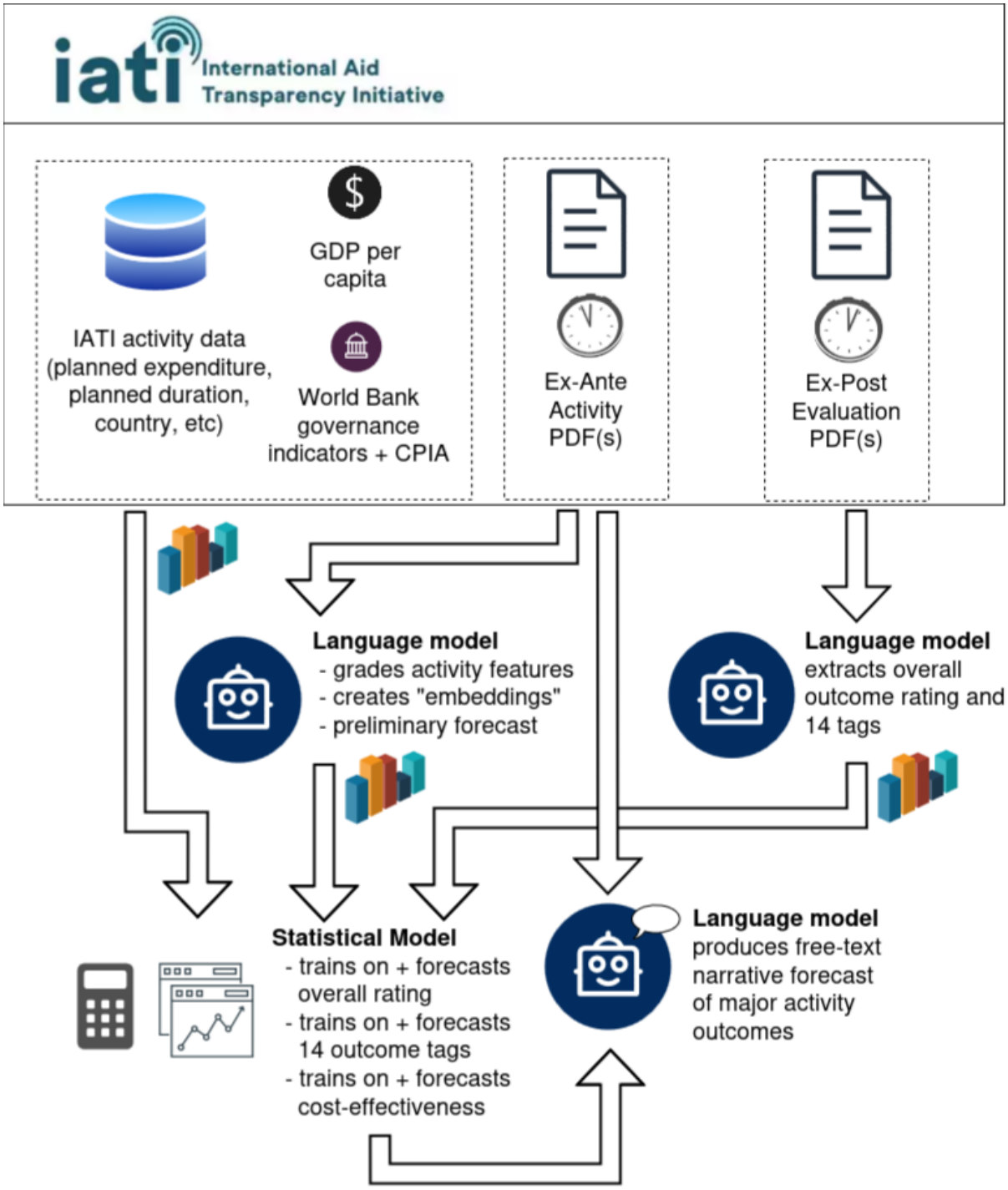

Figure 5: Overview of the forecasting pipeline. Features are produced using IATI activity data and external by-country datasources (GDP per capita, CPIA scores, and World Bank governance indicators). More features come from pre-activity PDFs processed by *gemini-2.5-flash*. *gpt-3.5-turbo* generated more features by grading them directly, and *gemini-embedding-001* created more features using semantic embeddings. Next, *deepseek-R1* produced a preliminary forecast for overall ratings and outcome tags. A statistical model also trains on and forecasts overall ratings, 14 outcome tags, and cost-effectiveness, optionally using the *deepseek-R1* forecasts as input features. *gemini-2.5-flash* processes ex-post evaluation PDFs to extract the rating and 14 tags, which serve as training targets for the statistical models. Lastly, *deepseek-R1* receives the outputs of the statistical model and the ex-post extraction to produce a free-text narrative forecast of major activity outcomes, graded by *gemini-2.5-flash-lite*.

3 Results

Results are organized by first listing the temporal information leakage considerations, followed by forecasting results on the test set for all models. See Section 3.7 for a summary of the results.

3.1 Temporal Information Leakage

The test set had a high rate of temporal leakage for LLM forecasts. Reading a few forecasts from *deepseek-R1* (minimal prompt) revealed several clear cases of leakage in the first few activities inspected. Reading the LLM forecasts from the same model in the validation set did not reveal any temporal leakage. To investigate more systematically, a custom python script searching for common keywords such as COVID or "Ukraine war" where that event occurred after the activity start date found that 3% of *deepseek-R1* forecasts in the validation set had COVID leakage (all started 2015–2016, before Jan 2020), with no Ukraine war words or words relating to emerging conflicts or financial collapse. However, consistent with the findings from manual inspection, the script revealed that 34% of *deepseek-R1* LLM forecasts on the test set had leakage, with the majority regarding COVID. It is hypothesized that this is due to the high influence of COVID and general upheaval in 2019, 2020, and 2021, which especially affected activities starting between 2016 and 2020.

All LLM forecasts with leakage were removed from forecast skill evaluation. Even after these removals, the test set results averaging language model forecasts performed significantly better than the validation set. However, this should be considered untrustworthy and understood to have a clear unfair advantage compared to a real forecast or the validation set, as the script may have missed some number of other leakage-related strings.

Direct inspection of dozens of LLM-generated grades did not reveal any leakage. However, running the leakage detection script revealed 3% of the train and validation set and 5% of the test set activities had leakage in at least one of their grades from *gemini-2.5-flash*. These grades were reset to the median train+val value and treated as "missing" before any results were reported.

3.2 Forecasting Ratings with Statistical Models

Overall, the forecasting system developed is capable of forecasting evaluation ratings significantly above chance on out-of-time activities at 95% confidence, with the RF+ET achieving a pairwise ranking skill of 60 % [95% CI: 51 %, 66 %]. This can be contrasted with the R^2 on the test set of -0.009, although it is still well above the baseline R^2 of

-0.053 from the ridge regression.

Despite its low test set accuracy, the model performs comparably to published literature on the training and validation set. The full RF+ET model has an R^2 on the combined training and validation set of 0.32. Existing literature as of 2021 reports at most 30% unexplained variance for World Bank ratings using linear OLS models (Ashton et al., 2023). An OLS linear model trained on all features in the final 4 reporting organization dataset after applying the mode-demeaned fixed effect for the training and validation set shows an R^2 of 0.18 and an adjusted R^2 of 0.14. This is in the expected range given that no evaluation-time information, such as the final duration and expenditure of the program, was used. No comparable out-of-time, time-ordered split analysis was identified in the literature. As expected for forecasting under data distribution shift, results on the out-of-time validation set were significantly weaker. This is likely due to significant structural shift over time in evaluation rating practices and increased influence of external factors such as the COVID-19 pandemic in the test set period.

Several factors improved model performance. As the ET model performed similarly to the RF model with a slightly different algorithm, averaging the two also improved forecasting performance on the test set (with the exception of a few of the least forecastable tags, which were worsened by ensembling). Further successful methods are listed below.

Language Model Features Used for the Statistical Model Adding the language-model-derived features, including LLM grades (finance, integratedness, implementer_performance, targets, context, risks, and complexity), embedding features (UMAP X, Y, and Z), and sector cluster allocations to the RF raises test R^2 from -0.026 to -0.009 ($\Delta R^2 = 0.017$; see Table 2, rows "Random Forest, no LLM features" vs. "Random Forest only").

Table 2: Test set performance in forecasting ratings across forecasting methods. Rows are sorted by ascending R^2 (higher is better). RMSE and MAE are lower-is-better, others are better if higher. Bold indicates the best value in each metric column. MAE is often better when the metric was rounded, explaining why the org-mode MAE is best. This is because the activity success ratings are integers, but the statistical models produce continuous prediction. The "LLM forecast (ridge)" variants combine models using the 300 latest-starting activities in the validation set for calibration/combination using the ridge regression model. XGBoost was found to underperform the RF+ET when used as the LLM Forecast base. The "no LLM features" RF+ET excludes the following features: finance, integratedness, implementer_performance, targets, context, risks, complexity, umap3_x, umap3_y, umap3_z, sector_distance, country_distance, and the 15 sector allocation features. Note: rows shown in orange include LLM Forecast predictions, which were found to benefit from future leakage and should be treated cautiously.

Method	R^2 ↑	RMSE ↓	MAE ↓	Acc. ↑	WG Spear. ↑	WG Pair. Rank. ↑
RF+ET + LLM Forecast (simple avg)	**0.067**	**0.894**	0.662	0.460	**0.303**	**0.645**
RF+ET all features (no year corr)	-0.004	0.928	0.684	0.465	0.205	0.597
RF+ET all features + year corr	-0.009	0.930	0.704	**0.485**	0.205	0.597
XGBoost all features	-0.013	0.932	0.693	0.460	0.144	0.567
RF+ET + LLM Forecast (ridge)	-0.023	0.936	0.702	0.475	0.149	0.574
RF+ET, no LLM features	-0.026	0.938	0.680	0.475	0.070	0.541
RF+ET (default params)	-0.047	0.947	0.709	0.455	0.185	0.590
Ridge Baseline (risks + org only)	-0.053	0.950	0.695	0.465	0.004	0.507
Mode of reporting-org score baseline	-0.079	0.962	**0.658**	0.460		
Plain OLS GLM	-0.114	0.977	0.767	0.410	0.070	0.535

Explaining Feature Importances SHAP (SHapley Additive exPlanations) interprets model predictions by decomposing each prediction into a baseline (the mean model output) plus additive contributions from individual features.

SHAP analysis reveals that, beyond reporting organization and region, easier target achievability, favourable external context, and sufficient annual expenditure (on a log scale) correlate with activity success (see Figure 6).

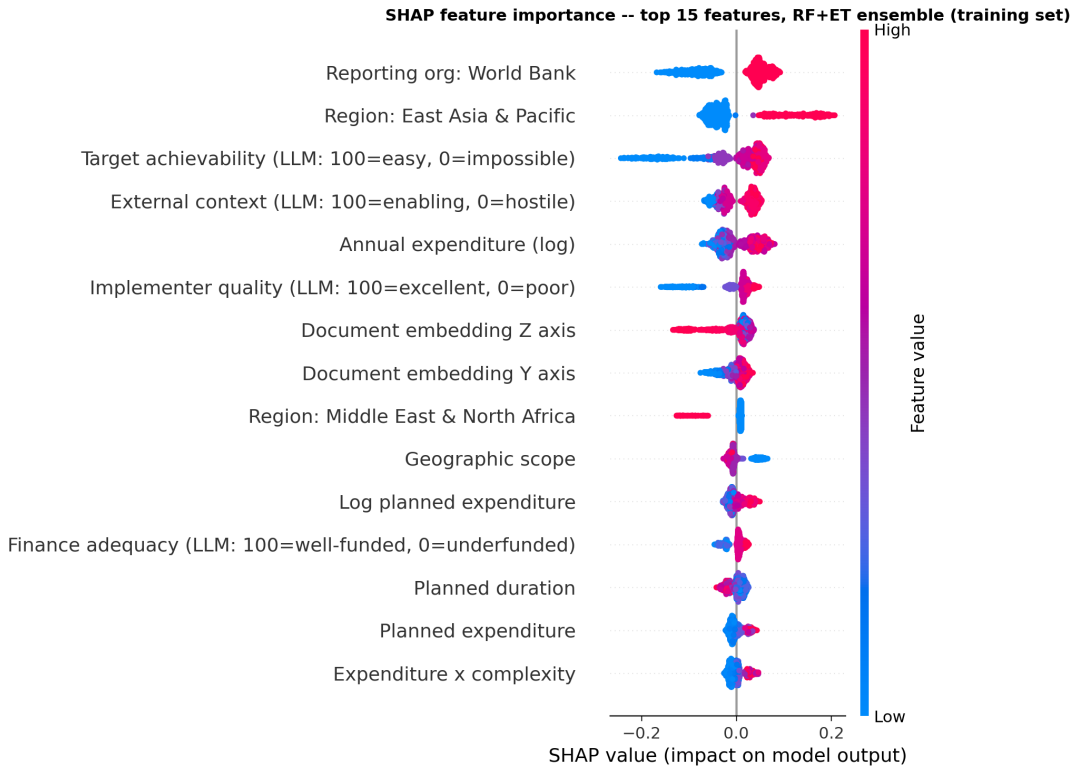

Figure 6: A SHAP analysis of ratings on the validation set from the RF. Red indicates an increase in the value of the feature, while blue indicates a below-average value. Points to the right of zero shift ratings up, points to the left of zero shift ratings down.

Table 3: Random-forest drop-one feature importance, sorted by absolute change in the prediction from the feature being increased by 1 standard deviation (Δ `pred_1sd`). Each row reports the decrease in validation R^2 and change in pairwise ranking fraction when the feature is removed (ΔR^2), along with the share of training rows that were median-imputed for that feature (`Miss` %). Negative ΔR^2 and ΔPairwise indicate improvements to forecasting on the test set if a feature is dropped.

Feature	Miss %	$\Delta R^2 \uparrow$	ΔPairwise ↑	Δpred_1sd
Target achievability (LLM: 100=easy, 0=impossible)	7.0	0.0100	0.0231	0.0878
Document embedding Z axis	8.5	0.0104	0.0139	-0.0666
External context (LLM: 100=enabling, 0=hostile)	6.5	0.0060	0.0089	0.0640
Sector: capacity building	47.0	0.0039	0.0032	-0.0448
Region: East Asia & Pacific	0.0	0.0135	0.0334	0.0404
Implementer quality (LLM: 100=excellent, 0=poor)	5.5	0.0050	0.0089	0.0339
Annual expenditure (log)	8.5	0.0006	-0.0080	0.0289
Document embedding Y axis	8.5	0.0036	0.0067	0.0236
Planned duration	0.5	-0.0003	0.0029	-0.0268
Expenditure x complexity	11.0	0.0037	0.0037	0.0231
Reporting org: World Bank	0.0	0.0301	-0.0057	0.0225
Geographic scope	0.0	0.0034	-0.0091	-0.0191
Region: Middle East & North Africa	0.0	0.0015	0.0076	-0.0174
Finance adequacy (LLM: 100=well-funded, 0=under-funded)	9.0	0.0087	0.0160	0.0149
Document embedding X axis	8.5	0.0068	0.0117	-0.0126
Country governance quality (CPIA)	31.5	0.0025	0.0005	0.0124
Sector: contingencies	47.0	0.0044	0.0031	0.0124
Region: Europe & Central Asia	0.0	0.0031	0.0009	0.0098
Finance type: loan	0.0	0.0026	0.0058	-0.0089
Log planned expenditure	5.5	0.0025	-0.0028	-0.0002
GDP per capita	7.5	0.0047	0.0005	0.0086
Sector: project management	47.0	0.0053	0.0079	0.0074
Region: Latin America & Caribbean	0.0	0.0044	0.0039	-0.0068
sector clusters missing	0.0	0.0037	-0.0003	-0.0062
sector cluster increased managed forest land	47.0	0.0051	0.0056	-0.0053

3.3 Forecasting Ratings and Narrative Forecasts with LLMs

Figure 7 summarises performance across all LLM forecasting configurations. The best narrative similarity grade is 88.19 (a "B+", corresponding to forecasts that are on average between "mostly accurate" and "highly accurate"), achieved by the RF-forced method with full context (RAG+KNN+S1+S2). The minimal-prompt baseline achieves the highest R^2 for rating prediction but the worst narrative similarity, while RF-forced methods dominate on both narrative grade and R^2 probability.

One research question of this thesis is how differing methods of LLM forecasting apply in the context of forecasting international aid success ratings. More generally capable models were found to be better forecasters. The LLM's ability to predict ratings did not clearly correlate with narrative quality. In fact, the best ranking skill from an LLM was associated with the least accurate stated reasons. The strategy of incorporating additional information relevant to the forecasting question as well as references to similar activities does not clearly improve forecasting skill.

Inspection did not reveal significant information leakage on the validation set, but considerations of the impacts of COVID and other events were prevalent in the test set. Indeed, a test of *gpt-3.5-turbo* and *gpt-4* with the same prompt as *deepseek-R1* revealed that the absolute error increased significantly on average after the model cutoff date relative to the RF+ET model. While *gemini-3-pro-preview* and *deepseek-R1* with thinking enabled performed better than the RF+ET ensemble on within-group pairwise ranking, this may be due to increased information leakage from LLM forecasts. Future work should test these more powerful models against larger ensembles of results after their cutoff dates. This would provide a fair test against statistical models which were at a lower overall risk of information leakage, as their training process was directly controllable.

Narrative Grades Narrative similarity between forecasts and summaries of post-activity evaluations was usually correlated with improved rating prediction, with the exception that very low-context prompts can forecast accurately but for the wrong reasons (the "Baseline" method) (see Figure 7). When the predicted rating from the best statistical model is injected into the prompt, LLM forecasts become much more similar to the post-activity evaluation summaries. This provides evidence that while language models can accurately forecast the reasons activities would succeed or fail if given the evaluation ratings, they generally lack the ability to correctly weigh the various factors to produce a calibrated forecast.

Rating accuracy alone should not serve as a benchmark for LLMs, given that the prompt with the worst narrative similarity also had the highest forecasting success. This may reflect improved model ability when the context window is less full: the model forecasts better when not overwhelmed with information, but the absence of that information

impairs its ability to identify the correct reasons for its forecasts. Alternatively, the model may focus on what it does know, such as supply chain disruptions or conflict during that period in the host country (potentially introducing more temporal leakage).

While ratings are not clearly correlated with better similarity grades, RAG, KNN retrieval, and extra reasoning of the models (stages 1 and 2) consistently improve narrative textual similarity to the post-activity summary, albeit by a small margin. Simply presenting the grading model with the *gemini-2.5-flash* summary of risks of the project from the project documents led to a forecasting score of 80.69 (between a "C+" and a "B-", corresponding to an "Adequate" forecast, with only "Adequate", rather than a good, focus on the drivers). However, the risks did not typically include any affirmative forecasts of post-activity outcomes, which may in part explain the low ratings. The best forecasting method in terms of similarity to the post-activity document summary was the forced ratings with a large amount of context (RAG+KNN) and included stage 1 and 2, achieving a grade of 88.19. This corresponds to a "B+" grade given by *gemini-2.5-flash-lite* (sitting midway between "Good" and "Excellent", meaning the forecasts were on average between "mostly accurate" and "highly accurate", with at least one major outcome forecast, and at least one key driver of the outcome identified on average). Without any additional methods, *deepseek-R1* can typically achieve a "B" grade on average, while use of the statistical model or of *gemini-3-pro-preview* can lift scores to the "B+" range. Overall ranking skill also improves significantly.

Fine Tuning The training loss for *gemini-2.5-flash* steadily decreased over fine-tuning, falling by 95% by the end of the process. This was tested on an early subset of the validation data which excluded BMZ activities. The performance results were mixed. While the R^2 performance improved modestly, it remained worse than simply guessing the mean rating in the validation set, and remained far below the performance of the more powerful *gemini-3-pro-preview*. Although the DPO objective is specifically intended to improve within-group pairwise ranking, fine-tuning slightly worsened it. Fine-tuning 20 epochs for 50 pairs cost approximately 140 euros. Scaling up to more training points does not appear to be a promising strategy for reaching the performance of *gemini-3-pro-preview* or the combined statistical models.

RAG and KNN Adding RAG and KNN produced small differences in rating forecasting skill. However, narrative similarity scores generally improved on the validation set when RAG and KNN context was included. This indicates that while the LLM identifies more relevant information when reasoning about the final forecast, it is insufficiently calibrated to translate that additional information into improved predictions.

Forecasting LLM Model Comparison When compared directly on the validation set using the same prompt, *gemini-3-pro-preview* performs better than *deepseek-R1* on all metrics, mirroring results from early tests on a different set of activities. However,

language models exhibit high forecasting variability across models and prompts, making comparisons more difficult.

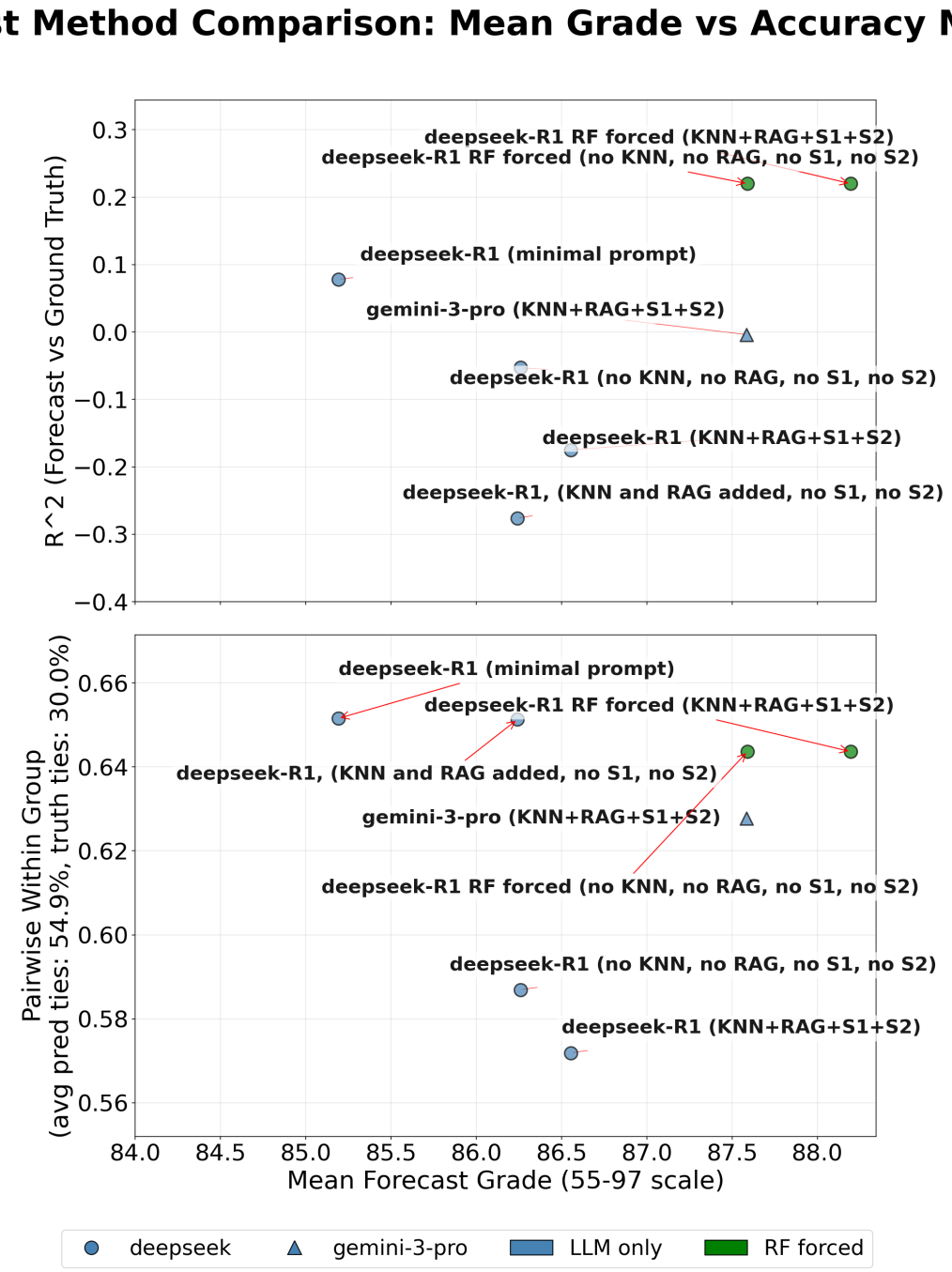

Figure 7: A comparison of LLM forecasting methods for the validation set. The x-axis shows the mean narrative forecast grade on a scale from 55 to 97. The y-axis in the top row shows R^2 between predicted and true ratings (negative values indicate worse-than-mean predictions). The y-axis in the bottom row shows within-group pairwise ranking (higher is better; 0.5 = chance). The main finding is that RF-forced methods (green) dominate on narrative grade and R^2, while the minimal-prompt baseline achieves the highest within-group pairwise ranking, illustrating a tension between ranking skill and narrative quality. RF refers to the RF+ET ensemble with all features and year correlation added, trained on the training set only. 288 of the 300 validation set activities with overall ratings had sufficiently detailed outcome evaluations to produce a summary. "KNN" retrieves 3 similar examples into the prompt, "RAG" additionally retrieves information the LLM searched for in pre-activity documents, "S1" refers to additional reasoning injected into the context about why the activity outcome may be overall unsatisfactory or moderately satisfactory, "S2" refers to additional reasoning in the context about why the activity outcome may be overall satisfactory.

3.4 Forecasting Cost-Effectiveness

For activities marked as dollar-per-unit, dividing by each activity's total expenditure reported by IATI and examining forecast aggregate cost-effectiveness outcomes yielded neither R^2 above 0 nor within-group pairwise ranking above 50%.

Cost-effectiveness outcomes were found to require more work to extract sufficient data for reliable outcome prediction. Furthermore, forecasting cost-effectiveness was found to be more challenging than forecasting overall success ratings, in contrast to the claims from prior literature (Goldemberg, Jordan, and Kenyon, 2025). One finding was that new water or sanitation piping connections is a more clearly comparable outcome, although there may be systematic differences in the costs of sanitation connections and water supply that are not disambiguated by the model.

It was also found that failing to divide by the total disbursement as marked in IATI increases predictability. This is in part because there is noise in the total expenditure, and partly because when not dividing by expenditure, the model learns to predict linearly higher cost-effectiveness outcomes correlating with expenditure.

Overall, little can be concluded from the cost-effectiveness outcome analysis. Expert coding is likely required for robust extraction of quantitative outcomes, and funding breakdowns for projects should be used to more accurately evaluate cost-effectiveness, rather than the coarse assumption that all funding for a project is evenly split among non-overlapping outcomes. New methods or increasing the number of activities analyzed for cost-effectiveness should be explored in future work.

3.5 Forecasting Outcome Tags

Outcome tags provide a richer window into project execution than a scalar rating alone. This section reports results from the statistical tag predictors trained on the same feature set as the rating forecasting models.

Consistent Correlation of Outcome Tags with Overall Ratings A natural validation of the outcome tags as meaningful measures of project performance is that they should correlate with overall evaluator ratings in the expected direction. This is apparently consistent with the extracted binary ratings. Tags representing good project execution correlate positively with overall ratings. Targets Met or Exceeded shows the strongest positive Pearson correlation with ratings ($r = +0.3930$), followed by High Beneficiary Satisfaction or Reach ($r = +0.3416$), Private Sector Engagement ($r = +0.3200$), Policy and Regulatory Reforms ($r = +0.3172$), Capacity Building Delivered ($r = +0.2655$), and Improved Financial Performance ($r = +0.1566$). Tags representing project difficulties are negatively correlated: Activities Not Completed shows the strongest negative correlation

($r = -0.2177$) and Implementation Delays shows a meaningful negative correlation ($r = -0.0944$). Project Restructured ($r = +0.0183$) and Targets Revised ($r = -0.0525$) are not meaningfully correlated with the overall rating in either direction.

Overall Ratings are Predictive of Tag Outcomes When the overall evaluation rating is incorporated as an additional feature alongside the pre-activity predictors, all 14 predictable tags improve on all evaluation metrics in Table 4. This provides evidence that the evaluation rating carries information beyond what is captured by project documents alone, strongly contrasting with the finding of (Goldemberg, Jordan, and Kenyon, 2025) that ratings do not aid outcome prediction.

Test-Set Performance To assess generalisation, all tag predictors were evaluated on the test set activities. Approximately 45% of within-group pairs had differing tag outcomes and therefore contributed to pairwise ranking; the remaining were tied in the true label and were excluded. Across the 14 RF+ExtraTrees tag models evaluated on the test set, the mean within-group ranking skill is 60 % (range: 48 %–77 %) averaging both evaluated methods on the test set.

While the ensembling with ET slightly decreased average scores for low-performing tags on the test set, the more forecastable tags saw improved forecasting when ET was averaged. Similarly, the overall success rating forecast was slightly improved when ET was ensembled with the RF on the test set.

As a secondary accuracy metric, the weighted accuracy across all 14 tags on the test set (for the uniform feature and tree parameter set method) improved from a majority-class baseline of 62.6 % to 67.9 %, an improvement of 5.3 percentage points. Weights are proportional to the number of test-set activities per tag. Per-tag forecasting performance on the test set is shown in Table 4.

3.6 Skill Decline with Increased Gap from Train to Test

Models from this work can only serve as decision support tools for new activities if they maintain above-chance forecasting skill despite large gaps between training and test activity start dates. Given the average ∼6 year duration of activities and the need for large training sets, forecasting new activities requires models to maintain performance with a ∼10 year gap between training and testing. A 450 activity rolling window training set was constructed, where the gap between the mean start date trained on and tested against the test set was varied. Performance declines on R^2 were consistently ∼-0.03 per year, while within-group pairwise ranking skill declined at a rate of ∼0.24% per year. For an activity starting in 2026, the expected decline corresponds to approximately an additional 6-year gap beyond the existing train-test gap; performance should therefore drop by approximately -0.18 in R^2 and -1.4% in within-group pairwise ranking. Within-group

Table 4: Outcome tag prediction results on the test set. $p < 0.05$ determined via permutation sampling on the test set. Consistent features shown are those in the top 10 by |SHAP|, with the same sign, and at least 5 % of total importance, in every split of 3 equal random training splits (RF and ET averaged). Features are ordered by mean importance across splits; features absent from any split's top 10 or exhibiting a sign flip are omitted. $\overrightarrow{\text{Green}}$ and $\overleftarrow{\text{Red}}$ indicate positive and negative relationships respectively. Tmp. Corr.: temporal correction in units of 1/100 %/yr. Feature abbreviations: $U_x/U_y/U_z$ = UMAP semantic embedding coordinates; dist = similarity distance group (sector + country distance combined); dur = planned duration; exp = planned expenditure group (raw/log/log-per-year combined); scp = activity scope; WB = World Bank reporting-org fixed effect. Groups and tags sorted by accuracy vs. baseline descending.

Tag (shown if $p < 0.05$)	WG Pair. Prob.	WG Spear.	Brier	Brier Base.	Acc.	Acc. Base.	Frac. True (train, test)	N_{test}	Tmp. Corr.	Consistent Features
Activity Rescoping										
Closing Date Extended*	0.77	0.44	0.184	0.258	0.73	0.53	0.53, 0.40	186	-0.65	$\overrightarrow{\text{exp}}$, $\overrightarrow{\text{scp}}$
Project Restructured*	0.75	0.43	0.198	0.267	0.69	0.60	0.60, 0.46	186	-0.22	$\overrightarrow{\text{WB}}$, $\overrightarrow{\text{exp}}$, $\overrightarrow{\text{scp}}$, $\overrightarrow{\text{dur}}$
Targets Revised*	0.63	0.21	0.226	0.259	0.61	0.58	0.58, 0.48	186	+0.13	$\overrightarrow{\text{WB}}$, $\overrightarrow{\text{exp}}$, $\overrightarrow{\text{scp}}$
Finance & Budget										
Funds Cancelled or Unutilized*	0.68	0.28	0.217	0.265	0.67	0.60	0.60, 0.48	186	-0.35	$\overrightarrow{\text{scp}}$, $\overrightarrow{\text{exp}}$
Improved Financial Performance	0.56	0.09	0.243	0.250	0.59	0.51	0.51, 0.48	186	+0.02	—
High Disbursement	0.56	0.11	0.243	0.247	0.58	0.57	0.43, 0.44	186	None	$\overleftarrow{\text{exp}}$
Over Budget*	0.66	0.24	0.223	0.232	0.67	0.56	0.44, 0.32	63	-0.50	$\overleftarrow{U_z}$, $\overleftarrow{\text{dist}}$, $\overleftarrow{\text{exp}}$
Funds Reallocated*	0.64	0.24	0.214	0.222	0.67	0.60	0.40, 0.31	186	+0.05	$\overleftarrow{\text{WB}}$, $\overleftarrow{\text{exp}}$, $\overleftarrow{\text{scp}}$
Target Achievement										
Targets Met or Exceeded*	0.60	0.17	0.244	0.253	0.55	0.58	0.42, 0.47	171	+0.13	$\overleftarrow{\text{exp}}$
Capacity Building Delivered	0.59	0.12	0.122	0.123	0.86	0.84	0.84, 0.86	146	+0.47	$\overleftarrow{\text{exp}}$
Gender-Equitable Outcomes	0.48	-0.04	0.193	0.193	0.75	0.80	0.80, 0.75	106	+0.39	—
High Beneficiary Satisfaction or Reach	0.53	0.04	0.156	0.155	0.81	0.79	0.79, 0.81	146	+0.14	—
Policy and Regulatory Reforms	0.57	0.09	0.176	0.177	0.78	0.72	0.72, 0.78	138	+0.53	—
Private Sector Engagement	0.51	0.01	0.233	0.233	0.64	0.59	0.59, 0.64	81	+0.34	$\overrightarrow{\text{dist}}$, $\overrightarrow{\text{exp}}$

pairwise ranking remains a favorable metric despite the long train-test gap, but R^2 drops well below what could be hoped for the model to recover. This reinforces the conclusion that absolute accuracy metrics should be avoided for forecasting evaluation, and that decision support with algorithms should focus on similar-start date portfolio selection.

3.7 Results Summary

The sections above present results for four outcome measures: overall evaluation ratings (key results in Table 2), LLM forecasts for overall ratings and narrative forecast similarity grades (key results in Figure 7), aggregate cost-effectiveness z-scores, and structured outcome tags (key results in Table 4).

Activity Ratings Results Overall rating results are shown in Table 2. The overall activity success rating prediction on the test set revealed an inability to explain variation in evaluation ratings in absolute terms on the test set. By contrast the within-group pairwise ranking ability on the test set was well above chance and statistically significant above 95% confidence. The use of LLM features improved all metrics, as did the hyperparameter tuning of the RF+ET model parameters. It was also revealed that nonlinear methods are essential for within-group pairwise ranking skill, with the OLS model both less accurate than the baseline and failing to rank within-group pairs.

Not all experimental methods improved the model. Inspection of the LLM forecasts on the test set revealed significant future leakage in the LLM's stated reasoning, so LLM-forecasting results on the test set should be understood as having an unfair forecasting advantage. Furthermore, the ridge regression method to calibrate LLM forecasts backfired on the test set, performing worse than a simple average of the LLM forecast with the RF+ET prediction. The temporal linear regression produced mixed results, decreasing overall accuracy, but improving within-group pairwise ranking and within-group Spearman ranking ability.

LLM Prediction of Overall Ratings LLM predictions on the test set were more accurate and had better within-group ranking skill than the statistical model, but were discovered to be strongly affected by future leakage, likely still improving from leakage despite attempts to remove instances of leakage from the forecasts. The validation set results did not demonstrate high rates of future leakage, and performed worse than statistical model predictions. A fusion of statistical models and LLM forecasting methods was found to generate the highest quality narrative predictions. The highest rating accuracy for LLM forecasts simultaneously produced the least accurate narratives as shown in Figure 7.

Cost-Effectiveness Activity ratings show only weak correlation with activity aggregate cost-effectiveness (Pearson $r = 0.0688$, computed across 533 activities). Among the tested

methods, the 3x weight on the last 20% of activities modestly improved performance on the test set. However, cost-effectiveness prediction was not shown to be statistically better than chance with the 95% permutation sampling for any scoring metric. The cost-effectiveness aggregate z-score is limited by inconsistency in its component indicators across activities, difficulty of interpretation, and low sample counts, indicating a need for a larger training set or a less noisy data collection method for international aid cost-effectiveness forecasting.

Outcome Tags Outcome tag results are shown in Table 4. Outcome tags had a similar mean within-group ranking skill as the overall rating model, with some notably stronger performing outcome tags. The per-tag feature selection and per-tag tree parameter tuning for outcome tags overall decreased forecasting skill compared to a uniform feature and tree parameter selection on the test set. Results are reported for the uniform tree parameter and feature set method rather than the hand-selected feature sets from the validation set. The simpler method of a uniform feature and tree parameter set is more likely to generalize forecasting skill to future activities and thus was given more detailed reporting in the results section.

Across 14 binary outcome tags, including activity rescoping, finance and budget, and target achievement, the RF+ET average with no per-tag feature selection achieved an average within-group pairwise ranking of 61% (range: 48% to 77%) for the 45% of pairs with differing outcomes, slightly outperforming the alternative model which used feature selection. By comparison, the per-tag feature selection model achieved an average of 60% (range: 48% to 77%).

4 Discussion

The direction of this work evolved significantly over the course of research. LLMs generally performed worse than the statistical machine learning models. Changes in forecasting methods tended to produce inconsistent performance changes across models and activity sets, making it difficult to compare methods or draw reliable conclusions about genuine improvements. Even with the very early cutoff date models *gpt-3.5-turbo* and *gpt-4*, there was still overlap between the training data and the start date, making any strong forecasting metrics likely in part due to leakage. This worry was validated by the discovery of specific examples on the test set where future information about COVID-19 and conflict outbreaks were used by the model to reason about overall success ratings. A further indication was the notable overall decrease in model accuracy for *gpt-3.5-turbo* and *gpt-4* when activities concluded their respective training cutoff dates, an accuracy decline which was not observed with statistical models.

These challenges with language models led the research to shift towards the use of statistical

models as the core experimental method. Language models were relegated to gathering relevant data and information about the activities and provide grades and embeddings to the machine learning model. Future methods should be developed to ensure leakage-free language model benchmarks such as the dynamic forecasting benchmark for geopolitical forecasting (Karger et al., 2024), or datasets with sufficient numbers of activities after the training cutoff.

Another major change in the direction of the research was in the choice of scoring metrics. A major finding was that within-group pairwise ranking was easier to forecast, more stable between splits, and more stable across varying start-date gaps. Furthermore, the key decisions between evaluations in practice are those of a portfolio manager, who works within a single organization and makes decisions for activities with similar starting dates. The question of which activities will perform well relative to each other is more important in that context than the absolute rating of each activity. Finally, it was observed that within-group pairwise ranking was a much more reliable metric than absolute accuracy for out-of-time generalization to the test set.

Lastly, the addition of outcome tags as a novel method for characterizing and forecasting activity outcomes uncovered several remarkably predictable binary tag outcomes, including Closing Date Extended, Project Restructured, and Funds Cancelled or Unutilized. As a means of predicting less subjective outcomes than overall ratings, outcome tag forecasting proved to be a much more fruitful direction than forecasting activity cost-effectiveness. Outcome tags also allowed for more confident additions of methods to try on the test set, because the effects of any change could be seen on all of the 14 tags, decreasing the chance that a method adopted improves forecasting skill only by chance.

4.1 Limitations

Several methodological limitations remain.

Future Leakage While significant attention was paid to assessing and ameliorating future leakage risk, LLM grades used as features for statistical forecasts, or the LLM forecasts on the validation set used for final reporting of LLM forecasting skill in Section 3.3 may still inadvertently leak signals from post-activity information. In particular, the discovery of temporal leakage in the dataset raises a concern that more leakage is possible. While leakage is likely constrained to be minimal on the validation set and the test set grades given low detection rates in the manually inspected random samples, there is still a possibility that leakage is responsible for a large share of forecasting skill. More systematic search for forecast leakage via additional keywords or automated LLM search could address this concern, but this is left for future work. There is also a risk that temporal leakage occurs in the forecasts in ways that have not yet been considered.

A related limitation is that BMZ-reported pre-activity documents were not available. The LLM was therefore instructed to extract data from evaluation documents, minimizing temporal information leakage as much as possible. The test set demonstrated a performance improvement in forecasting BMZ outcomes (only 10% of test set activities), which may have to do with this limitation.

LLM Extraction Quality A second important limitation is the quality of single-shot extracted LLM data. It would be valuable to rerun the grading scheme multiple times and ensemble the results for improved accuracy; this was not attempted due to the high API cost involved. A separate re-run for outcome tag extraction focused on whether each outcome tag did or did not occur could improve label accuracy as well but was not attempted due to API costs. It is likely that there is systematic error regarding omission of unsigned tags mentions, leading to erroneous marking of many unsigned tags as false. This issue could be addressed with per-tag scans of the evaluation documents rather than extracting all tags from a single LLM summary. This per-tag outcome extraction approach was not attempted due to high API costs.

Limited Feature Set Despite the extensive use of information from pre-activity documents, much more information could have been extracted. There was limited experimentation with embeddings, embedding space optimization, and UMAP parameter sizes. Only the targets and pre-activity summary were embedded, though embeddings ended up being among the most important features for the statistical model. A straightforward extension would be to embed the full language model grading justifications from *gemini-2.5-flash*, which could provide useful features capturing information that cannot be summarized from a single grade. Other features such as the average word length in the pre-activity documents could have provided a proxy for the care with which the application was prepared. Legal and contractual documents were not used but could also have been incorporated. Lastly, features were selected for overall rating forecasting but were used for outcome tags, cost-effectiveness, and LLM prompting. Additional features specific to the tags or cost-effectiveness outcome categories could improve forecasting skill.

Risks of Overly Trusting the Model Models trained for outcome tags, narrative forecasts, and overall rating tags should improve decision making about which activity proposals to prioritize in a portfolio, if adopted. However, there are some risks. First, users of predictive models may ascribe undue objectivity to algorithmic outputs, leading to "automation bias" (Logg, Minson, and Moore, 2019). This can cause decision-makers to overweight model recommendations relative to other sources of expertise. Another risk of incorporating algorithmic decision making into decisions is that grant officers may avoid pursuing projects that appear unlikely to succeed according to the model, narrowing the range of activities explored to those similar to historically successful ones. This may hinder adaptation to changing international aid contexts.

Limitations in Measuring Activity Success Only activities that received a formal evaluation appear in the dataset. Projects cancelled early, quietly wound down, or never subjected to review are absent entirely. The model therefore learns what distinguishes successful from unsuccessful activities among those deemed worth evaluating, which may not be the same as what distinguishes successful from unsuccessful activities in general.

A related conceptual limitation for rating forecasts is that the model forecasts evaluator ratings, not causal impact. A high rating reflects that an evaluator judged the activity to have met its objectives, which can be influenced by the quality of documentation, the ambition of the targets set, or the relationship between the implementing organisation and the evaluating body, independently of true effectiveness.

Other Limitations There are also risks that proposals could be gamed by proposing organizations to make their proposals better rated by the model. The usage of 59 separate features and multiple forecasted metrics ameliorates the concern. A last limitation is that no baseline from experienced development professionals making judgmental forecasts was available for comparison. The closest proxy is the ridge regression on the LLM-graded overall risk and reporting organisation (*Ridge Baseline (risks + org only)*), which achieved 51 % pairwise ranking on the test set suggesting that reading stated risks alone provides little forecasting power. A proper expert judgmental baseline would be a valuable addition in future work.

4.2 Extensions and Future Work

More Data A straightforward extension of this work is including the entire IATI dataset, rather than the ∼20% of the dataset pertaining to the environment and sustainability. An exponential saturation curve fit of the performance improvements and extrapolating the fit to the remaining IATI records suggest the 14 tags would have average improvements in within-group ranking probability from the current average of 60% to 64%, and overall ratings from the current 60% to 65%.

Improved Methods An untested method left for future work would be to use time-dependent coefficients on the nonlinear models, which would allow shifts in the influence of different input features to be captured and projected into the future as the relative importance of pre-activity input features changes over time.

Other than using the full dataset including activities unrelated to the environment and sustainability, extending the existing database using data from AidData could substantially increase the total number of activities, as AidData incorporates pre-activity and evaluation documents which are not reported directly to IATI.

Cross-Organization Applications The method is not limited to BMZ but could be applied to any organization with sufficient evaluation data. The overlap in effectiveness

data across organizations suggests that cross-organization models are feasible and beneficial: BMZ alone may lack enough labeled evaluations to train a reliable model, but the combined BMZ and World Bank dataset appears sufficient. Such a pooled model can transfer predictive signal from one reporting organization to another, improving forecasting performance for data-sparse reporting organizations.

Realtime Database Updating The method of data downloading and extraction could also be applied in real-time: the underlying database could update automatically as new activities are published to IATI. A real-time updated database could be used to gather a larger quantity of pre-activity documents and to gather guaranteed leakage-free data from the start of the activity, at a time when URL links have just been established and are generally still working, and pre-activity documents cannot have been covertly updated post-activity. Forecasting models in this case could be evaluated annually. Once a sufficient volume of new evaluation data accumulates after a model's training cutoff, the argument against temporal leakage becomes considerably stronger, providing a cleaner basis for assessing true out-of-sample performance.

New Methods for AI Forecasting As AI advances, the best way to use it shifts. At the beginning of this thesis in 2025, carefully designed prompting was the strategy with the best evidence, adopted by several academic works (e.g. (Halawi et al., 2024) (Kaiser et al., 2025)). By mid-2025, it became apparent that reinforcement learning on effective chain of thought from the models was making overly structured prompts ineffective. By 2026 the paradigm shifted again to agents that can explore documents freely, launch sub-agents, and train and run their own statistical models. Future work should explore the potential for agentic AI to improve forecasting skill beyond structured prompting as was attempted in this thesis. The best statistical forecasting models produced in this work can serve as a baseline against which more powerful forecasting systems using LLMs with significantly improved reasoning abilities can be compared.

Policy and Legislation The codebase and research presented here can be repurposed for other domains where impact forecasts are useful. A similar system with an expanded dataset could be used with little modification in domains such as public health, finance, and policy. Assessing the impact of policies before they are enacted is one clear use-case. Paired legislative and evaluation data already exist across multiple jurisdictions, though they have never been systematically linked for predictive modelling. The Congress.gov API provides real-time bill text and status for every bill since the 93rd Congress, with Government Accountability Office audits providing post-enactment reports and evaluations (Office, 2026). The EU Legislative Observatory (OEIL) tracks every proposal under the ordinary legislative procedure since 1994, with mandated REFIT evaluations providing ex-post assessments 4-5 years after entry into force (European Court of Auditors, 2018). The UK publishes Departmental Impact Assessments alongside every Government Bill, with Post-Legislative Scrutiny reports providing systematic follow-up (*Guide to Making*

Legislation (HTML) 2026). Finland publishes all Government Proposals (hallituksen esitykset) through the Eduskunta and Finlex systems prior to parliamentary consideration. Publicly available legislative impact assessments are provided by the Legislative Review Council (Lainsäädännön arviointineuvosto) via the Eduskunta portal (*Lainsäädännön Arviointineuvosto* 2026). In each case, the legislative and evaluation records are public; the gap is the absence of a linked dataset that pairs them and a decision support tool trained on early pairs of pre- and post-impact assessments and tested on later ones.

Futarchy One promising direction for future work is an application to Futarchy, first proposed by Robin Hanson (Hanson, 2013). The governing principle is "vote on values, bet on beliefs": democratic processes define a measurable welfare metric, while prediction markets determine which policies are expected to maximize it. For each policy proposal, two conditional markets are created, one paying out if the welfare metric improves when the policy is enacted and one if it improves when the policy is not enacted. The policy whose enactment market prices welfare more favorably is implemented. This aggregates dispersed information through financial incentives, separating the normative question of what to optimize from the empirical question of how best to achieve it.

The forecasting approach developed in this thesis could augment or partially replace the prediction market mechanism across the domains described in this section. In international aid, public policy, and legislation, the evidence databases already exist or are being assembled. Once built, statistical and LLM-based forecasts of activity outcomes are low-cost to generate and require no active market participants or financial liquidity. Rather than waiting for market prices to aggregate beliefs about which intervention will maximize a welfare metric, a forecasting model trained on historical outcomes could directly estimate the conditional expected welfare gain of each proposed policy. This lowers the barrier to Futarchy-style governance wherever liquid prediction markets are impractical to organize, extending its applicability from financial markets to any institutional context with a sufficient evidence base. While current methods in this thesis focus on providing a decision support tool, eventually with advanced AI capabilities and better benchmarks, beliefs about the likely impact of legislation may become scientific and objective, rather than political and subjective. This raises the prospect of more productive political debate, where empirical questions about policy effects are resolved by the forecasting system, leaving genuine value disagreements as the remaining source of contention.

5 Conclusion

5.1 LLM vs Statistical Models

Below is a summary of how statistical models and LLM forecasting methods compare. Both strategies were found to have strengths and weaknesses:

Strengths

LLM Forecasting

- Can explicitly reason and identify missing information
- General reasoning skills may transfer across domains
- Can produce narrative forecasts

Statistical Models

- Cheap and fast to iterate
- Can use large numbers of features without context limits
- Can incorporate LLM forecasts and grades as features
- Efficiently generalize over large datasets
- Easier to prevent future leakage
- Mature methodology

Weaknesses

LLM Forecasting

- Expensive and slow to iterate
- Difficult to calibrate
- Fine-tuning is expensive, and unavailable for best models
- Low interpretability and reliability
- Best models are closed-source
- Prone to temporal leakage
- Limited context window constrains attention and calibration

Statistical Models

- Cannot perform explicit reasoning
- Cannot directly use rich textual evidence or world knowledge
- More prone to overfitting dataset-specific quirks

5.2 Summary of Methods and Results

Table 2.13 summarises the key experimental methods tested in this thesis, with detailed descriptions below.

Statistical Model (Overall Rating) The baseline methods were a mode-of-organisation predictor and a ridge regression using only an LLM-generated risk grade and organisation identity. An RF alone was inferior to the RF+ET ensemble on the validation set (largely confirmed on the test set); RF+ET was therefore adopted as the experimental method. Experimental features added to the RF+ET ensemble included seven *gpt-3.5-turbo*-generated grades (financing quality, integratedness, implementer performance, contextual challenge, risk, complexity, and ease of targeted outcomes), three-dimensional UMAP embeddings of LLM-generated target text, sector and country distance features derived from those embeddings, and budget allocations clustered into 15 sectors. A nearest-neighbour weighted average of training ratings was also tested as a direct predictor on the validation set but did not improve over the mode baseline and was not assessed on the test

set. Hyperparameter tuning to restrict tree depth and leaf size improved generalisation on both validation and test sets. A linear ridge regression correction fitted on start year was applied to the ensemble output, with improvements on the validation set but mixed results on the test set. Combining the RF+ET forecast with the LLM forecast via simple averaging beat LLM adjustment via a calibrated ridge regression. XGBoost was also assessed on the validation set and test set and underperformed RF+ET in both cases.

LLM Forecast (Overall Rating and Narrative Quality) The baseline was a minimal single-stage prompt with activity metadata and a generated summary. Surprisingly, this was the best-performing method for overall ratings. Multi-stage prompting added two separate prior calls eliciting reasons the activity would go badly (S1) and reasons it would go well (S2), whose outputs were inserted into the final forecast context. KNN retrieval incorporated summarised mock forecasts of the three most semantically similar training activities, and RAG retrieval allowed the model to pull additional document excerpts it identified as relevant. Instructing the LLM to match the statistical model's predicted rating was the most effective single intervention for both narrative quality and rating accuracy. A fine-tuning attempt using Direct Preference Optimisation on *gemini-2.5-flash* over 50 training pairs produced mixed results and did not approach the performance of the larger *gemini-3-pro-preview* model. An LLM correction blend was also tested for outcome tag forecasting but consistently decreased performance and was not attempted on the test set.

Cost-Effectiveness The RF+ET statistical model was applied to cost-effectiveness z-scores, but validation performance did not exceed chance. No further methods were therefore pursued on the test set.

Outcome Tags The RF+ET ensemble with a shared set of 45 features was the core method. Additional methods validated on the training and validation sets and incorporated into the final tag models included a start-year linear trend correction, altering the default tree parameters, group averaging to stabilise predictions for low-data tags, and tag-dependent class-weight balancing. Per-tag feature selection and hyperparameter tuning improved validation performance but did not generalise to the test set. Ridge regression, XGBoost and LightGBM were assessed on the validation set and underperformed RF+ET on every tag.

Table 5: Summary of experimental methods by outcome target. ✓ improved on validation and test; × did not generalise to test; ? ambiguous or no clear improvement. LLM forecast rows: validation set only.

Outcome Target Are forecasts viable? (✓ / ? / ×)	**Variants of best method on validation set (test set performance)**	**Should variant be adopted?**
Statistical Model (Overall Rating) ✓ (better than chance with a 95% confidence using a permutation test)	LLM-generated features	✓
	Restricted tree complexity (RF+ET tuning)	✓
	Linear ridge regression adjustment to RF based on start year	?
	Simple average with LLM forecast	?
	LLM ridge adjustment of forecast	×
	XGBoost	×
LLM Forecast (Overall Rating) ? (did not outperform the RF+ET model on the validation set for rating, but performed well for narrative forecasts. Temporal leakage detected in test set, so test-set forecasts are not considered here.)	**Narrative quality (validation set):**	
	LLM instructed to forecast statistical model's rating	✓
	S1 (reasons would go badly) & S2 (reasons would go well)	✓
	KNN similar activities retrieval	✓
	RAG additional context retrieval	✓
	Rating accuracy (validation set):	
	LLM instructed to forecast statistical model's rating	✓
	S1 (reasons would go badly) & S2 (reasons would go well)	×
	KNN similar activities retrieval	×
	RAG additional context retrieval	×
Cost-Effectiveness ? (no methods perform better than chance in validation or test set using 95% permutation test)	No methods attempted for test set: validation and test set performance not above chance	
Outcome Tags ✓ (7/14 better than chance with 95% confidence using a permutation test)	Per-tag features & regularization	×

5.3 Answers to Research Questions

Each research question from the introduction is addressed below.

Can the overall rating, common true/false activity outcomes or activity cost-effectiveness be forecast from pre-activity information alone? Compared to the baseline of 51 % pairwise ranking for the risks extrapolation, the chosen statistical model with all input features reached 60 % [95% CI: 51 %, 66 %] on the test set for within-group pairwise ranking correctness. Across 14 binary outcome tags the RF+ET ensemble achieves above-chance within-group pairwise ranking on the test set, with a mean within-group pairwise ranking of 60% (range: 48%–77%), improving weighted accuracy over the majority-class baseline by 5.3% percentage points. The strongest-performing tags by both pairwise ranking and Brier skill were Closing Date Extended, Project Restructured, and Funds Cancelled or Unutilized. Activity cost-effectiveness could not be forecast above-chance.

How do judgemental forecasting methods compare to statistical models in forecasting international aid overall success ratings and common true/false outcome probabilities? LLM forecasts themselves (even when including a full RAG- and KNN-augmented multi-stage prompting pipeline) tend to underperform statistical models in this domain, while having the disadvantage of being costlier and more difficult to iterate on. While results were better on the test set, this is likely due to temporal leakage. However, LLMs provide meaningfully accurate narrative forecasts when combined with statistical models.

How do differing methods of combining language models and simpler statistical models compare in this domain? Embeddings of language models inserted into statistical models significantly improve forecasting ability. Furthermore, using embeddings as a means of selecting nearest neighbor mock forecasts and as a component of RAG retrieval reliably increased the graded similarity between LLM forecasts and the eventual outcome summary over all tested LLM forecast configurations. Language model embeddings were also effective as a method for clustering activity disbursements as measured by improved performance on the validation set.

Language models themselves were helpful in extracting grades for various aspects of the forecast, which improved the statistical model forecast. Among methods for directly blending LLM and statistical forecasts, simple averaging of the RF+ET and LLM predictions achieved the highest within-group pairwise ranking accuracy (64.5%), outperforming ridge-regression combination (57.4%) and the RF+ET alone (59.7%); however, the LLM forecasts used in this comparison were found to benefit from temporal information leakage, so this result should be interpreted cautiously. Directly prompting the LLM to agree with the RF model's predicted rating achieved the best narrative forecast quality without

relying on leakage-prone model outputs, at the cost of weaker independent ranking skill.

What methods improve the accuracy of narrative (qualitative) activity success forecasts in this domain? The experimental methods were successful in improving narrative forecast quality. These were: 1. to switch the model from the somewhat smaller *deepseek-R1* or *gemini-2.5-flash* models to the more expensive but more generally capable *gemini-3-pro-preview* model, 2. to incorporate additional retrieval of information via RAG and insert outcomes from similar past activities using the KNN method and 3. to directly prompt the model to come to the same conclusion as the RF model.

Additional prompts to elicit reasons the outcome may go well or badly (stages 1 and 2 in the methods) sometimes reduced scores in certain model configurations, although the best-performing models used these stages in their context window when forecasting. Similarly, fine-tuning sometimes helped and sometimes reduced scores.

What aspects of the activity available in the dataset at the beginning of the activity lead to higher or lower overall ratings or true/false outcome probabilities? As others in the literature have reported, decreased duration and especially increased planned expenditures tend to correlate with higher activity ratings (Vivalt, 2020) (Ashton et al., 2023) (Eilers et al., 2025). Rating organizations tend to differ systematically, and by-sector differences in ratings appear to be more significant than regional differences.

The planned expenditure, reporting organization and geographic scope consistently influenced many binary outcome tag forecasts.

Overall, feature importances are prone to shifting with new data, so more work should be done on model interpretation. One robust result is that most features contribute to accurate prediction, with no single feature or set of features dominating.

6 Data and Code Availability

Extracted data from IATI as well as LLM generated forecasts are available at `https://github.com/morganrivers/iati_extraction`. Code used to generate the results from the processed IATI data is available at `https://github.com/morganrivers/forecasting_thesis_code`. A decision support tool demonstrating the models developed during this thesis is hosted online at `https://forecasting-iati.onrender.com`.

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Declaration of AI Use

ChatGPT 5 and Claude Sonnet 4.6 were used occasionally to write individual sentences in the Methods section summarizing code and to format equations in LaTeX. All LLM-authored text was carefully checked for validity. ChatGPT 5.2 and Claude Sonnet were also used to provide feedback for spelling, writing style and grammar.

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